

Outline

- ❑ Introduction
- ❑ Thevenin's theorem
- ❑ Norton's theorem
- ❑ Numerical Examples

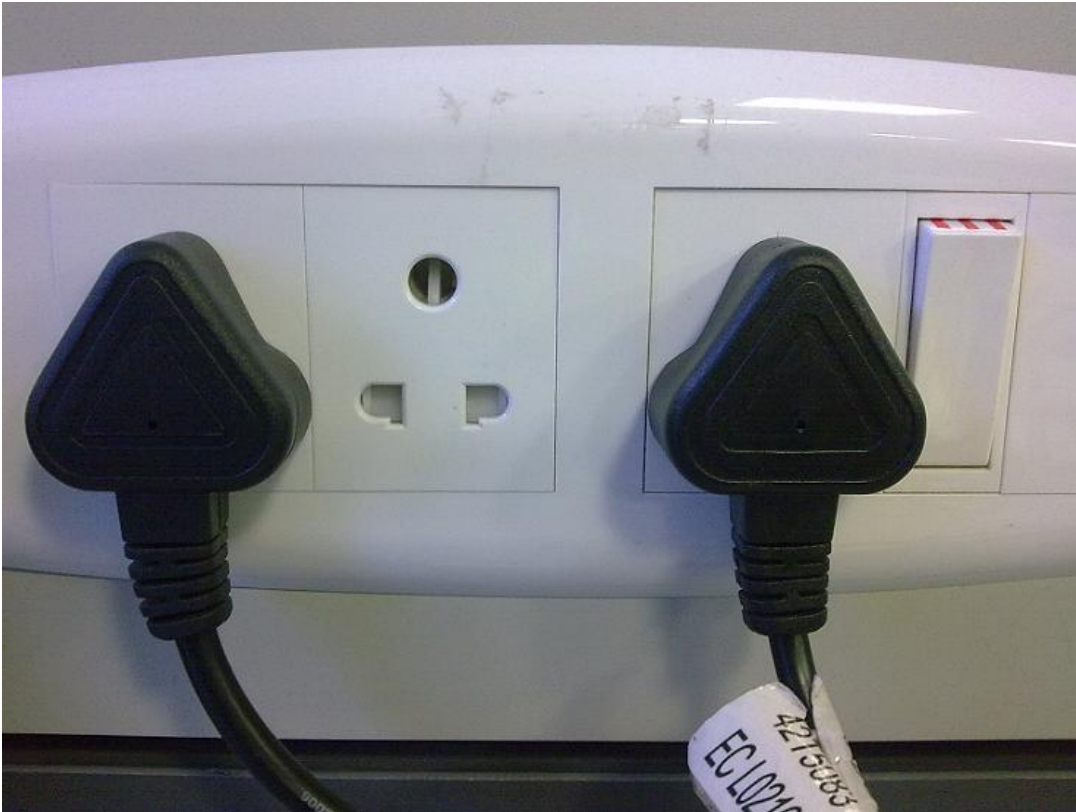
INTRODUCTION

Thevenin's Theorem Application

- It often occurs in practice that a particular element in a circuit is variable (usually called the load) while other elements are fixed.
- As a typical example, a household outlet terminal may be connected to different appliances constituting a variable load.
- Each time the variable element is changed, the entire circuit has to be analyzed all over again.
- To avoid this problem, Thevenin's theorem provides a technique by which the entire fixed part of the circuit is replaced by a very simple equivalent of a voltage source in series with an impedance.

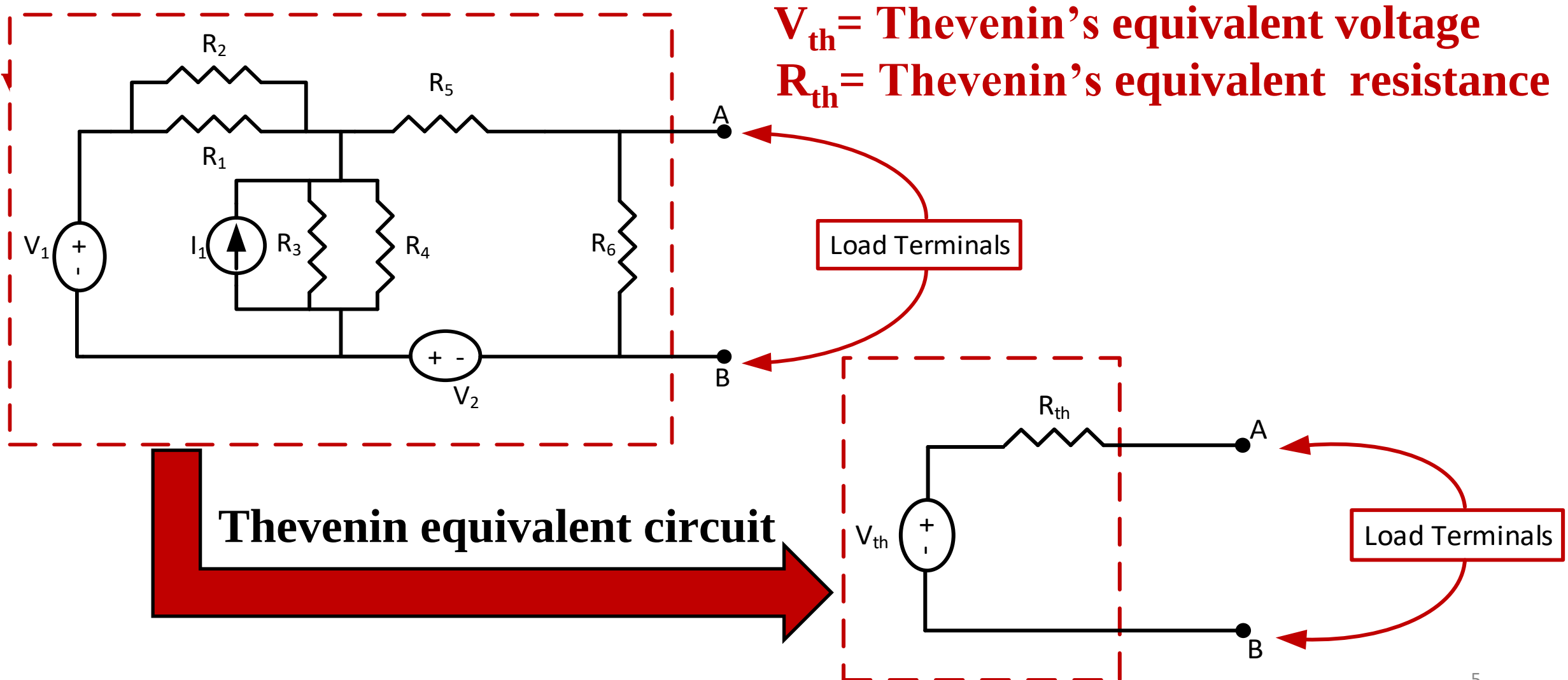
INTRODUCTION

Thevenin's Theorem Application Example



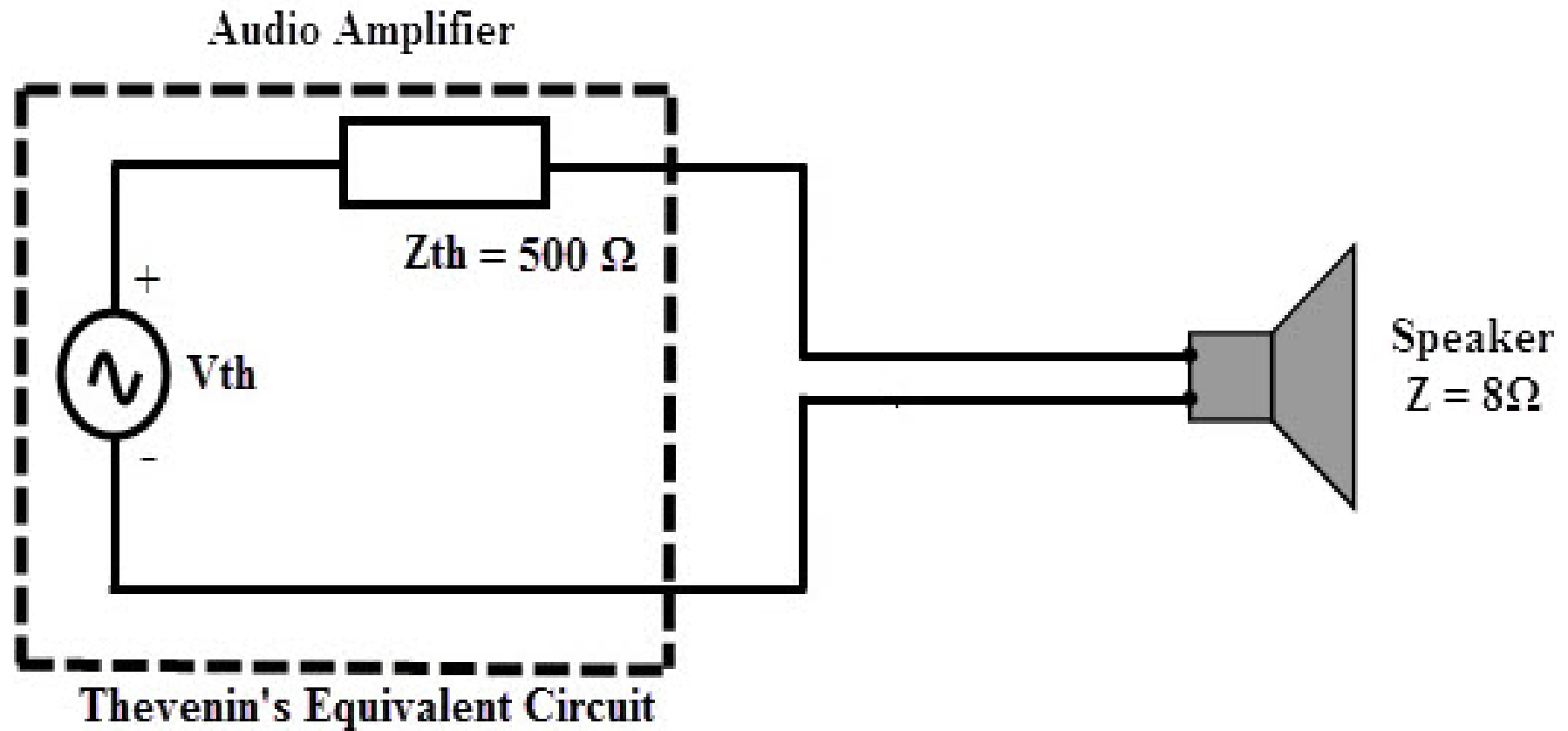
INTRODUCTION

Thevenin's Theorem



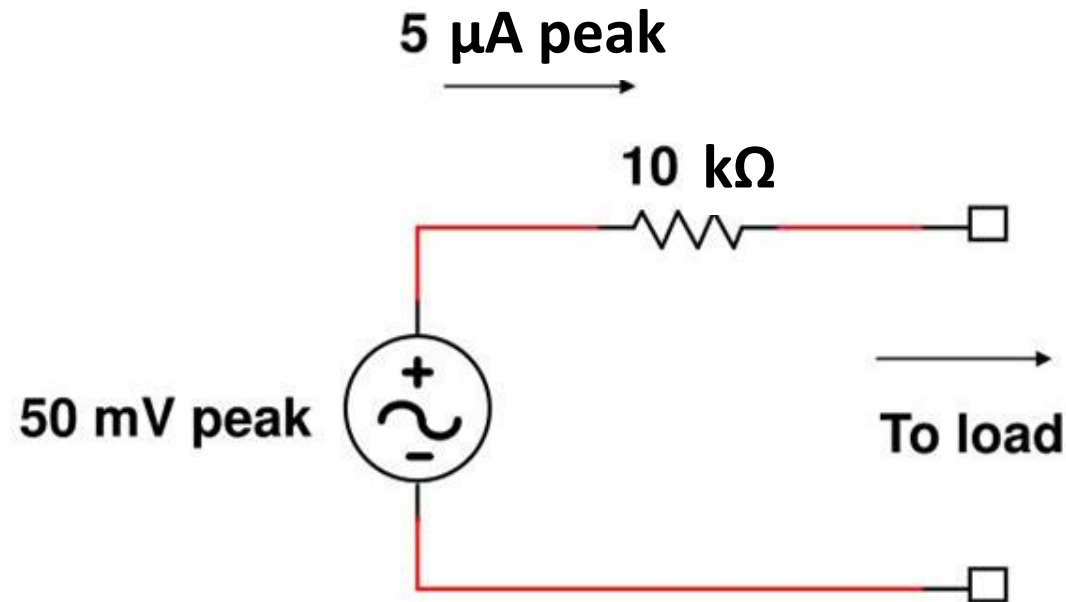
INTRODUCTION

Thevenin's Theorem application areas



INTRODUCTION

Thevenin's Theorem application areas



Microphone
10 kΩ output impedance
50 mV peak

THEVENIN'S THEOREM

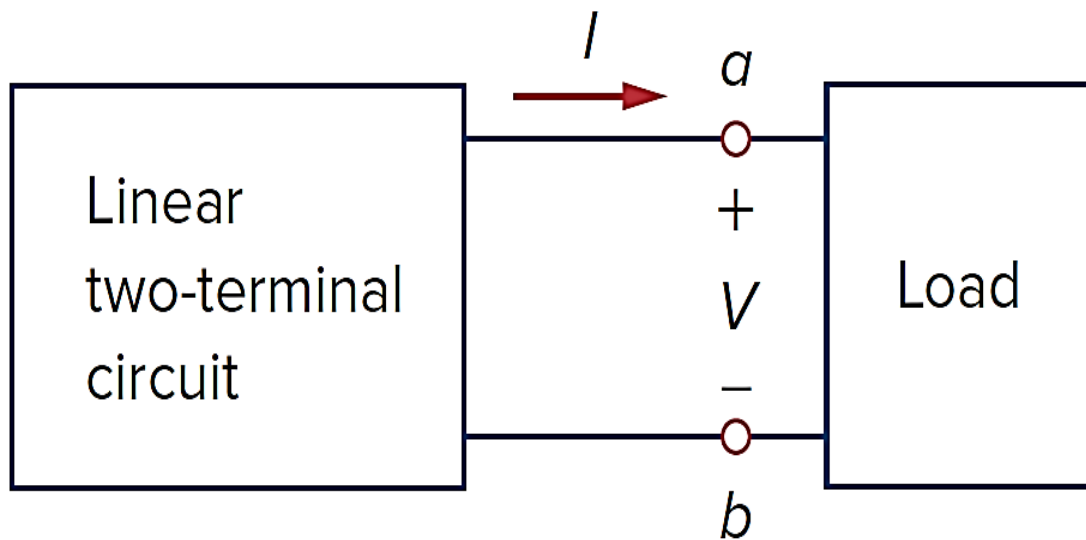
Statement: A linear two-terminal circuit can be replaced by an equivalent circuit consisting of a voltage source V_{Th} in series with a resistor R_{Th} , where V_{Th} is the open-circuit voltage at the terminals and R_{Th} is the input or equivalent resistance at the terminals when the independent sources are turned off.

Linear Circuit: A linear circuit is one whose output is linearly related (or directly proportional) to its input i.e. containing only linear elements eg. R, L, C, transformer

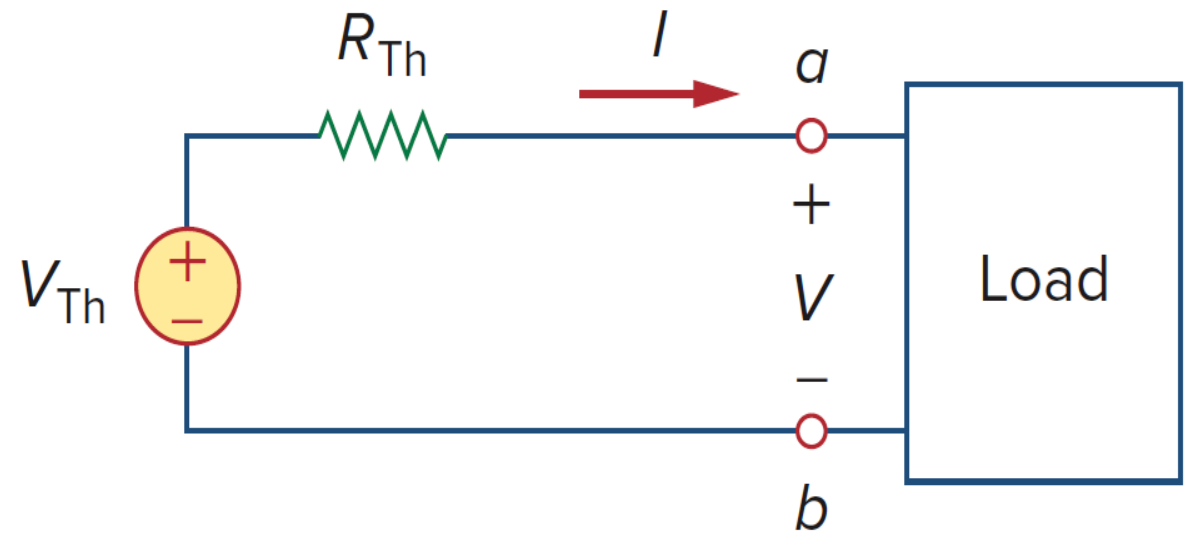
THEVENIN'S THEOREM

How to find Thevenin equivalent voltage V_{Th} and resistance R_{Th} ?

- The two circuits given below are said to be equivalent if they have the same voltage-current relation at their terminals.



(a) Original Circuit



(b) Thevenin equivalent circuit

THEVENIN'S THEOREM

- If the terminals $a-b$ are made open-circuited (by removing the load), no current flows, then the open circuit voltage across the terminals $a-b$ is equal to the voltage source V_{Th} .
- **Thus, V_{Th} is the open-circuit voltage across the terminal i.e. $V_{Th} = v_{oc}$**
- The input resistance (or equivalent resistance) of the dead circuit (all independent sources turned off) at the terminals $a-b$ in the Fig (a) must be equal to R_{Th} in Fig. (b) (input resistance with V_{Th} turned off).
- **Thus, R_{Th} is the input resistance at the terminals when the independent sources are turned off, i.e. $R_{Th} = R_{in}$**

THEVENIN'S THEOREM

Finding the Thevenin equivalent resistance R_{Th} :

Case 1: When the networks has no Dependent Sources

- Turn off all the independent sources, then R_{Th} is the input resistance of the network looking between terminals a and b .

Case 2: When the networks has Dependent Sources

- Turn off all independent sources, apply voltage source v_o across terminals a - b and determine the resulting current i_o . Then $\mathbf{R}_{Th} = v_o / i_o$.
- Alternatively, insert a current source i_o across terminals a - b and find the terminal voltage v_o . Again $\mathbf{R}_{Th} = v_o / i_o$.

THEVENIN'S THEOREM

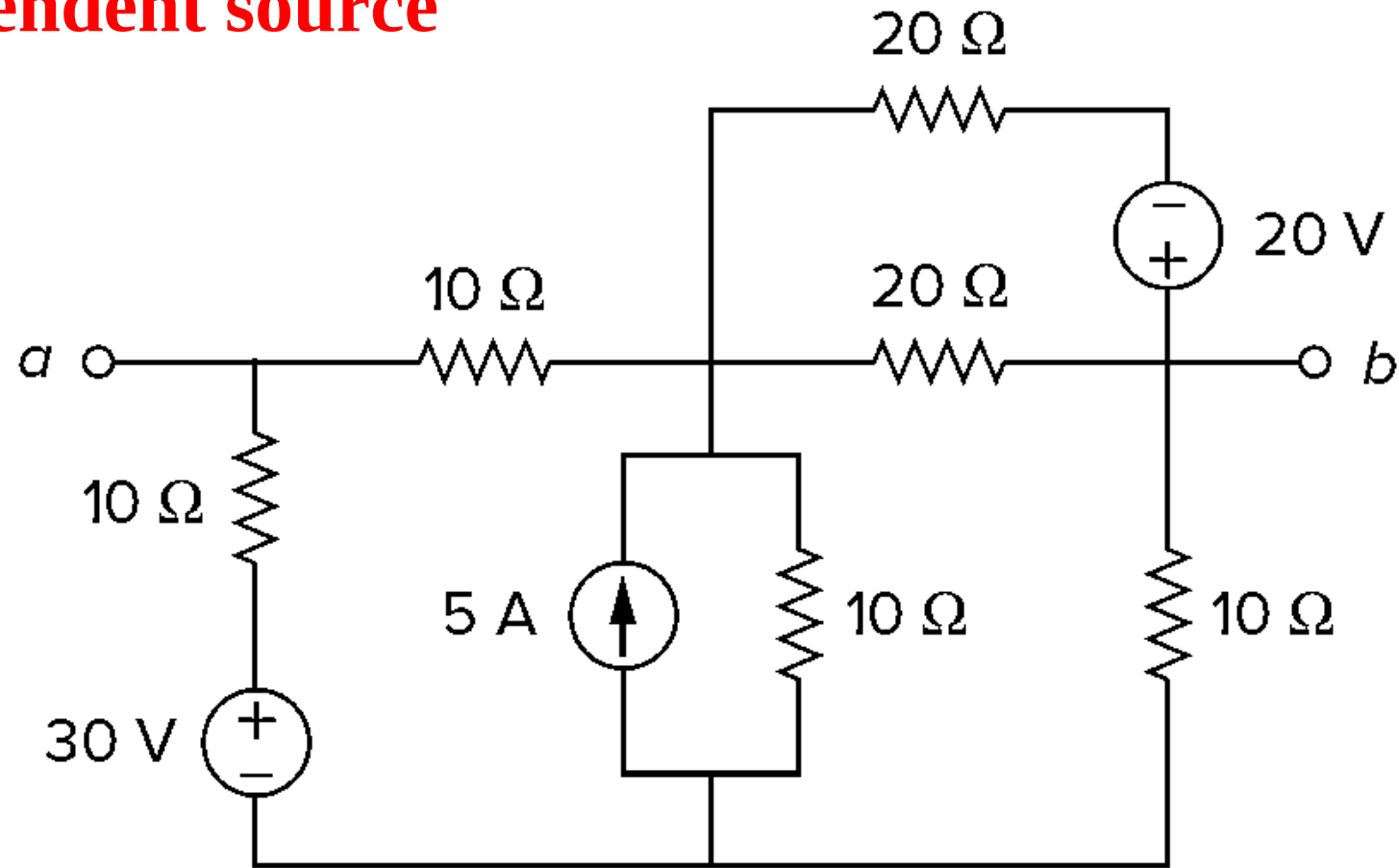
Steps to determine Thevenin's Equivalent Resistance (R_{TH}) and Voltage (V_{TH}):

- Remove load resistor R_L or any component connected across the terminals $a-b$ through which Thevenin equivalent is required.
- Determine R_{TH} by shorting all voltage sources and open-circuiting all current sources, and then calculating the circuit's total resistance as seen from the open terminals $a-b$.
- Determine V_{TH} by calculating the voltage between open terminals $a-b$ (by usual circuit analysis methods).

THEVENIN'S THEOREM

Example: To find Thevenin equivalent between terminals 'a' and 'b'

Case 1: No dependent source

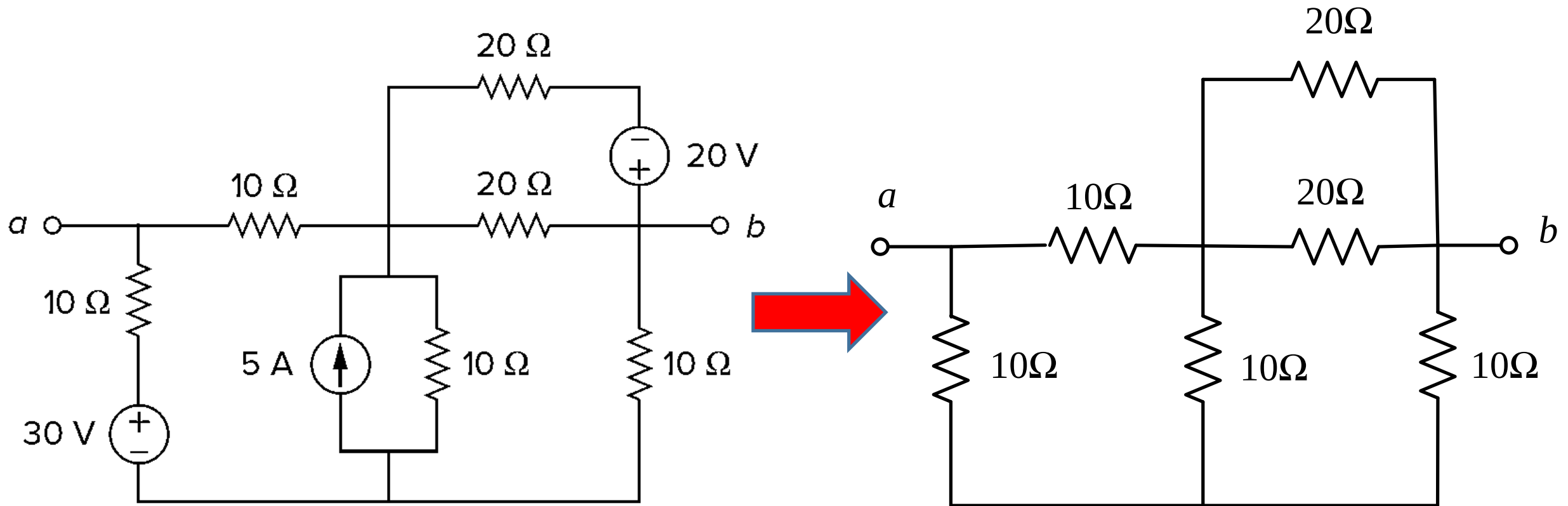


THEVENIN'S THEOREM

Step1: Remove all the independent sources.

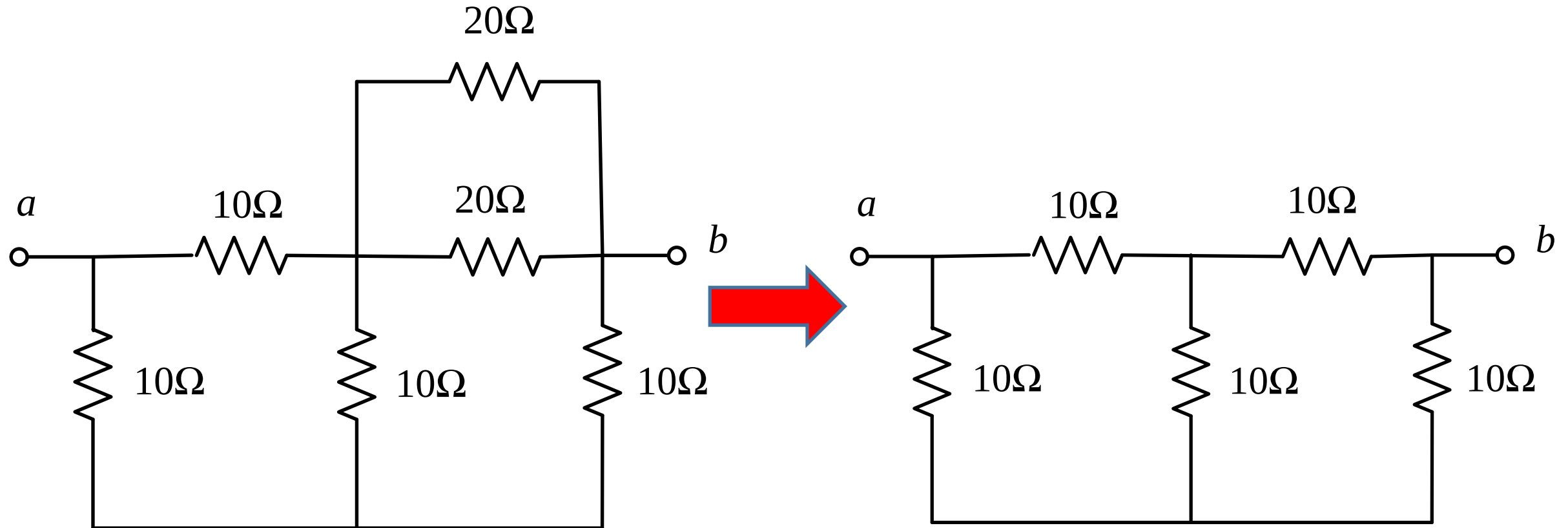
a) Replace voltage source by short circuit

b) Replace current source by open circuit



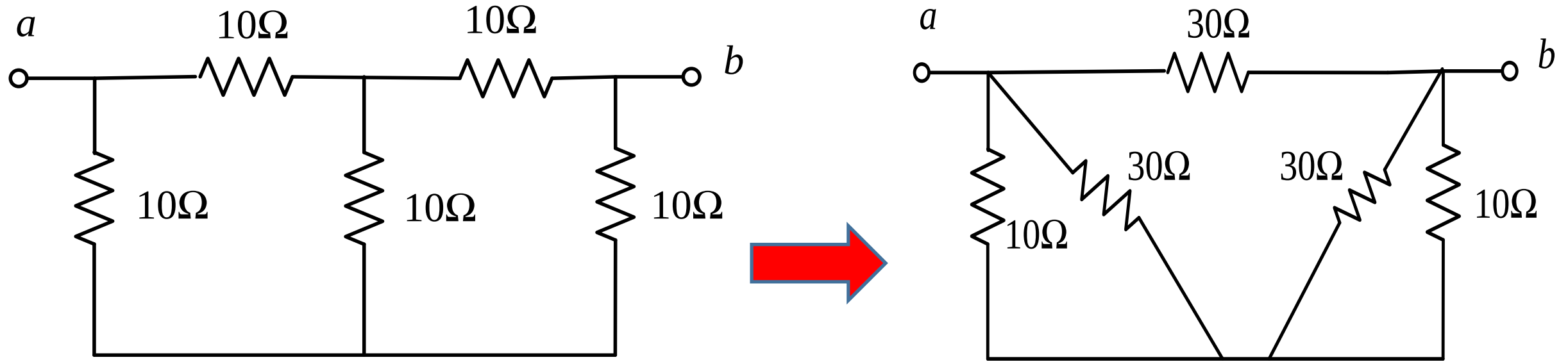
THEVENIN'S THEOREM

Step2: Finding R_{Th}



THEVENIN'S THEOREM

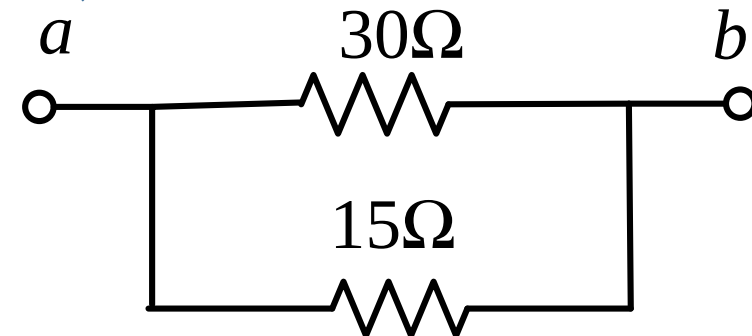
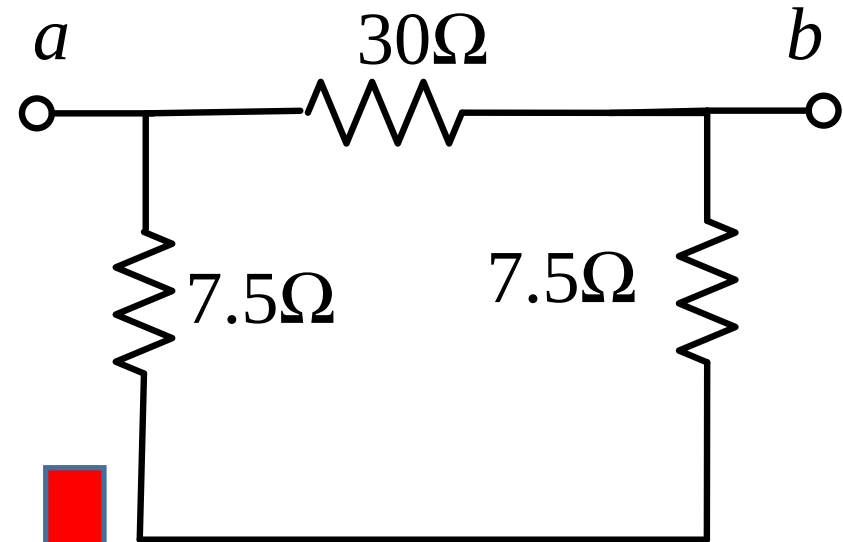
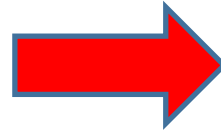
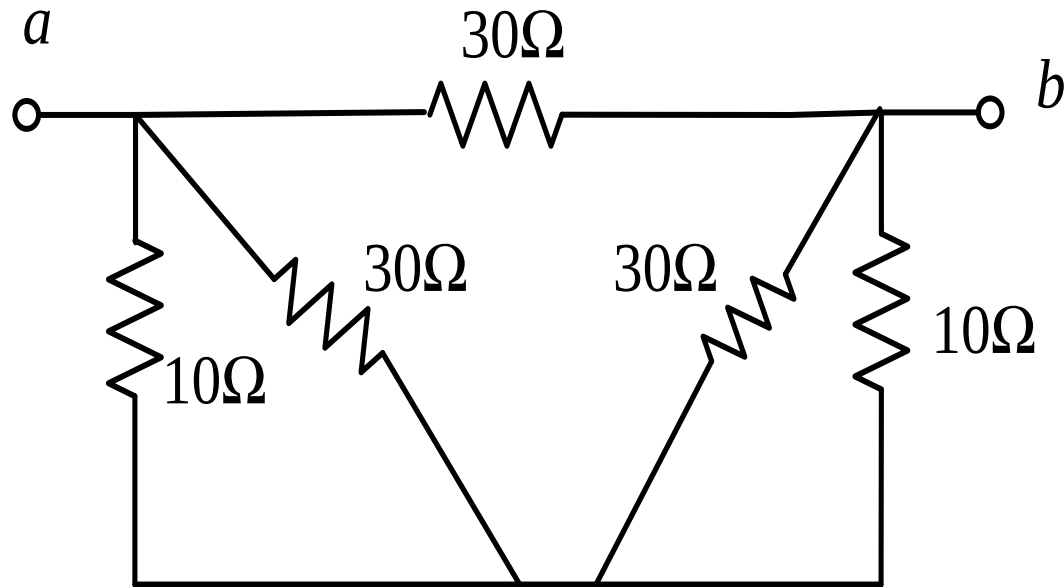
Step2: Finding R_{Th}



Star to Delta transformation

THEVENIN'S THEOREM

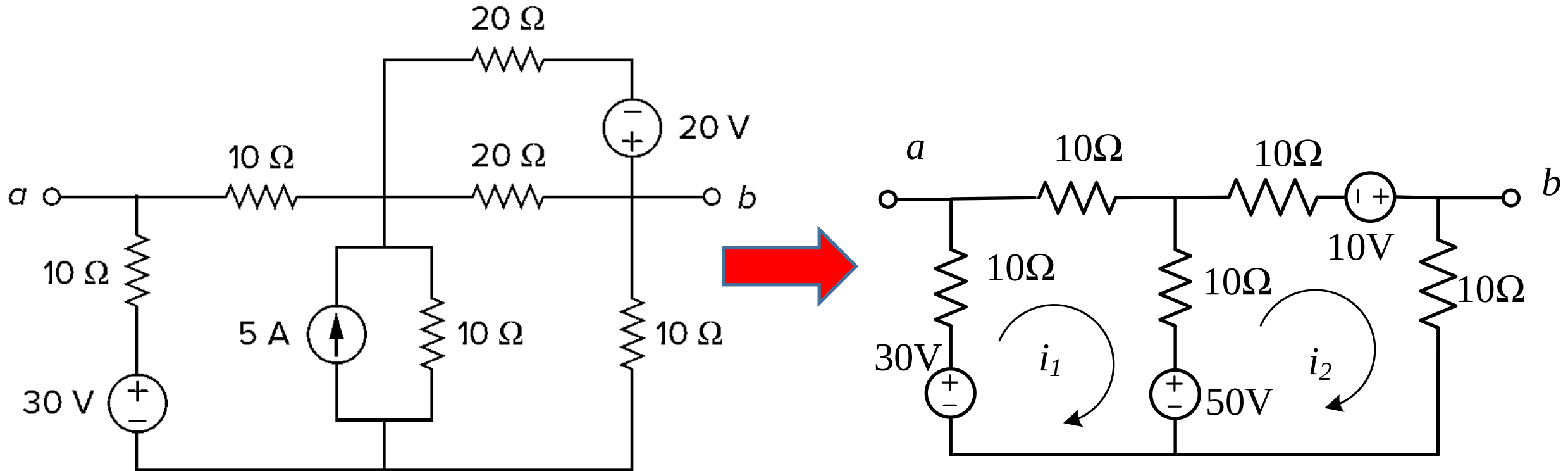
Step2: Finding R_{Th}



$$R_{Th} = 30 || 15 = 10\Omega$$

THEVENIN'S THEOREM

Step3: Finding V_{Th}



Source Transformations

THEVENIN'S THEOREM

Step3: Finding V_{Th}

For loop 1:

$$-30 + 50 + 30i_1 - 10i_2 = 0$$

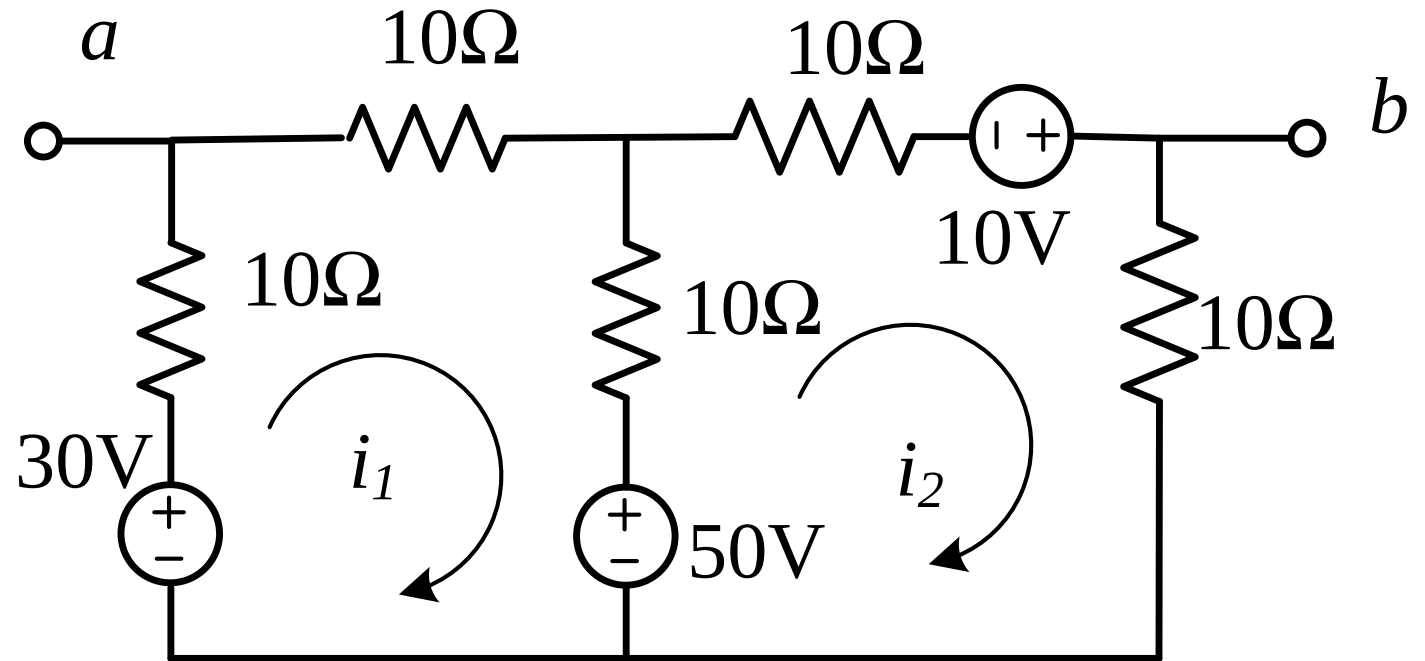
$$\Rightarrow -2 = 3i_1 - i_2 \quad (1)$$

For loop 2:

$$-50 - 10 + 30i_2 - 10i_1 = 0$$

$$\Rightarrow 6 = -i_1 + 3i_2 \quad (2)$$

On solving (1) and (2), $i_1 = 0$, $i_2 = 2A$



THEVENIN'S THEOREM

Step3: Finding V_{Th}

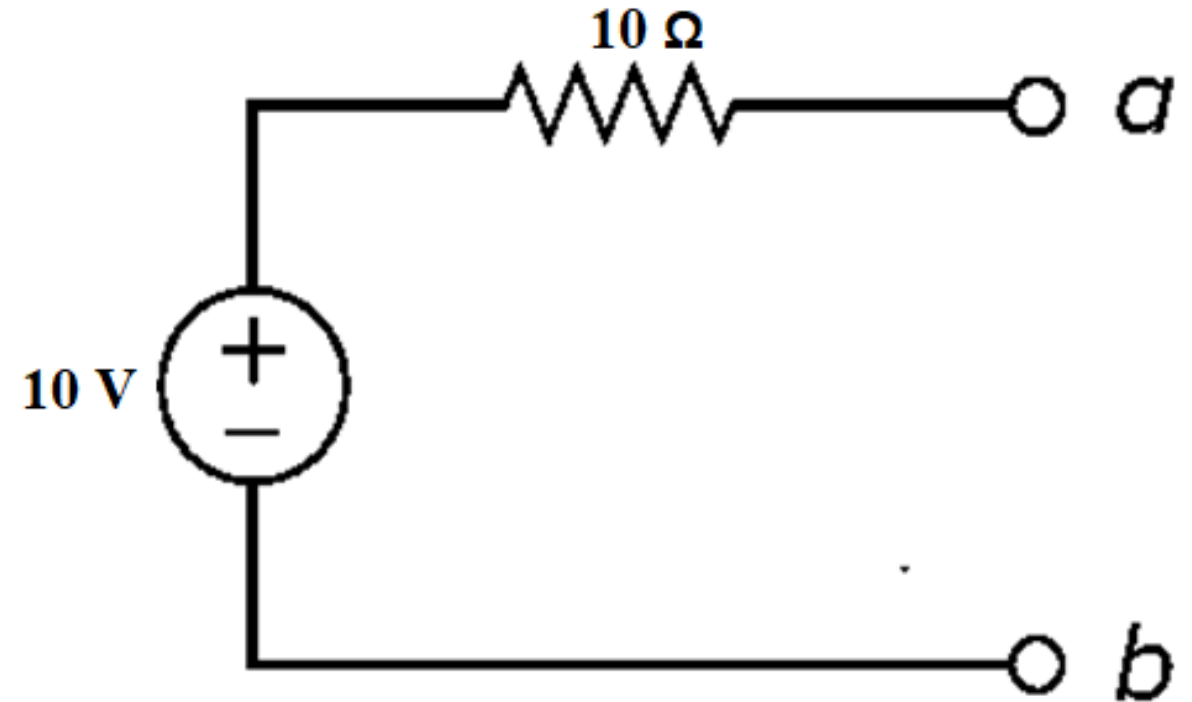
Applying KVL to the output loop,

$$-v_{ab} - 10i_1 + 30 - 10i_2 = 0$$

$$\Rightarrow v_{ab} = 10 \text{ V}$$

Therefore,

$$V_{Th} = v_{ab} = 10 \text{ V}$$

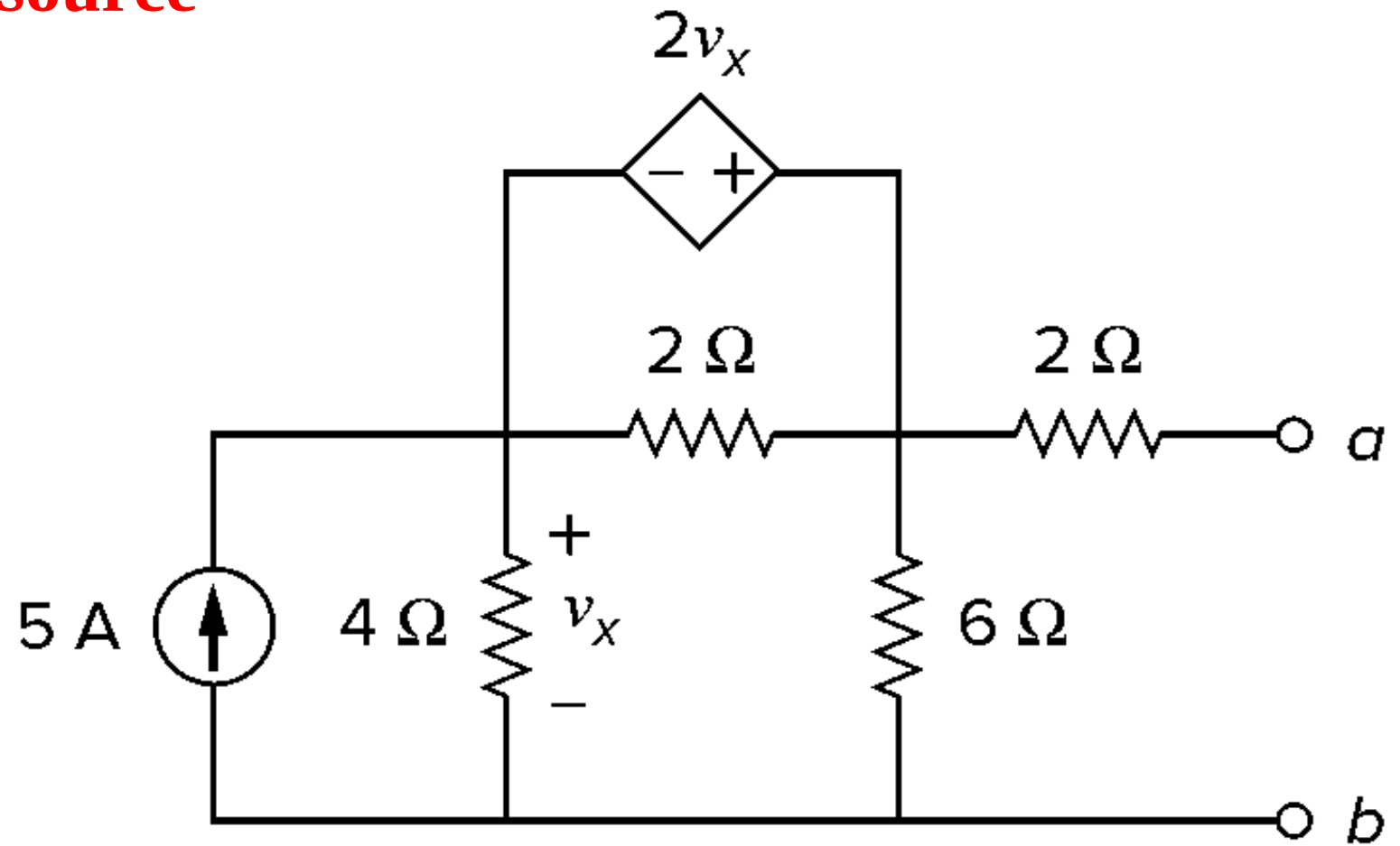


Thevenin Equivalent Circuit

THEVENIN'S THEOREM

Example: To find Thevenin equivalent between terminal 'a' and 'b'
Case 2: With dependent source

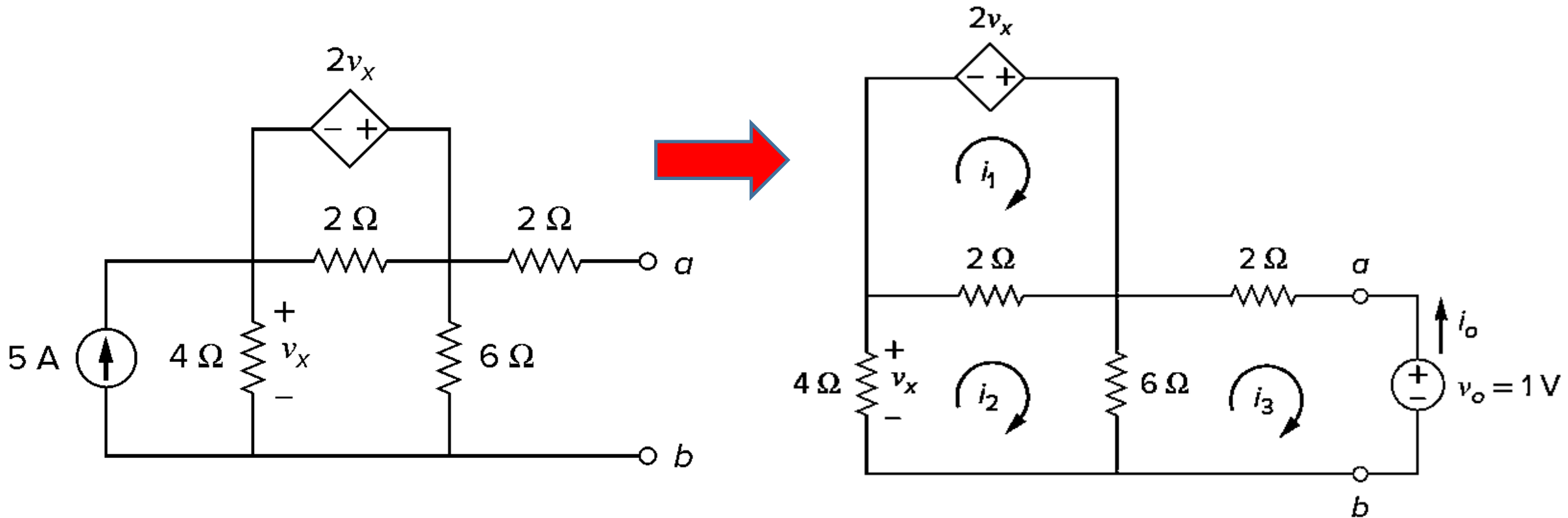
Turn off all independent sources, apply voltage source v_o at terminals **a** and **b** and determine the resulting current i_o . Then $R_{Th} = v_o/i_o$.



THEVENIN'S THEOREM

Step 1. Remove dependent sources

5A current source is replaced by open circuit and set $v_o = 1\text{V}$.



THEVENIN'S THEOREM

Step 2. Find R_{Th}

KVL for loop 1:

$$-2v_x + 2(i_1 - i_2) = 0$$

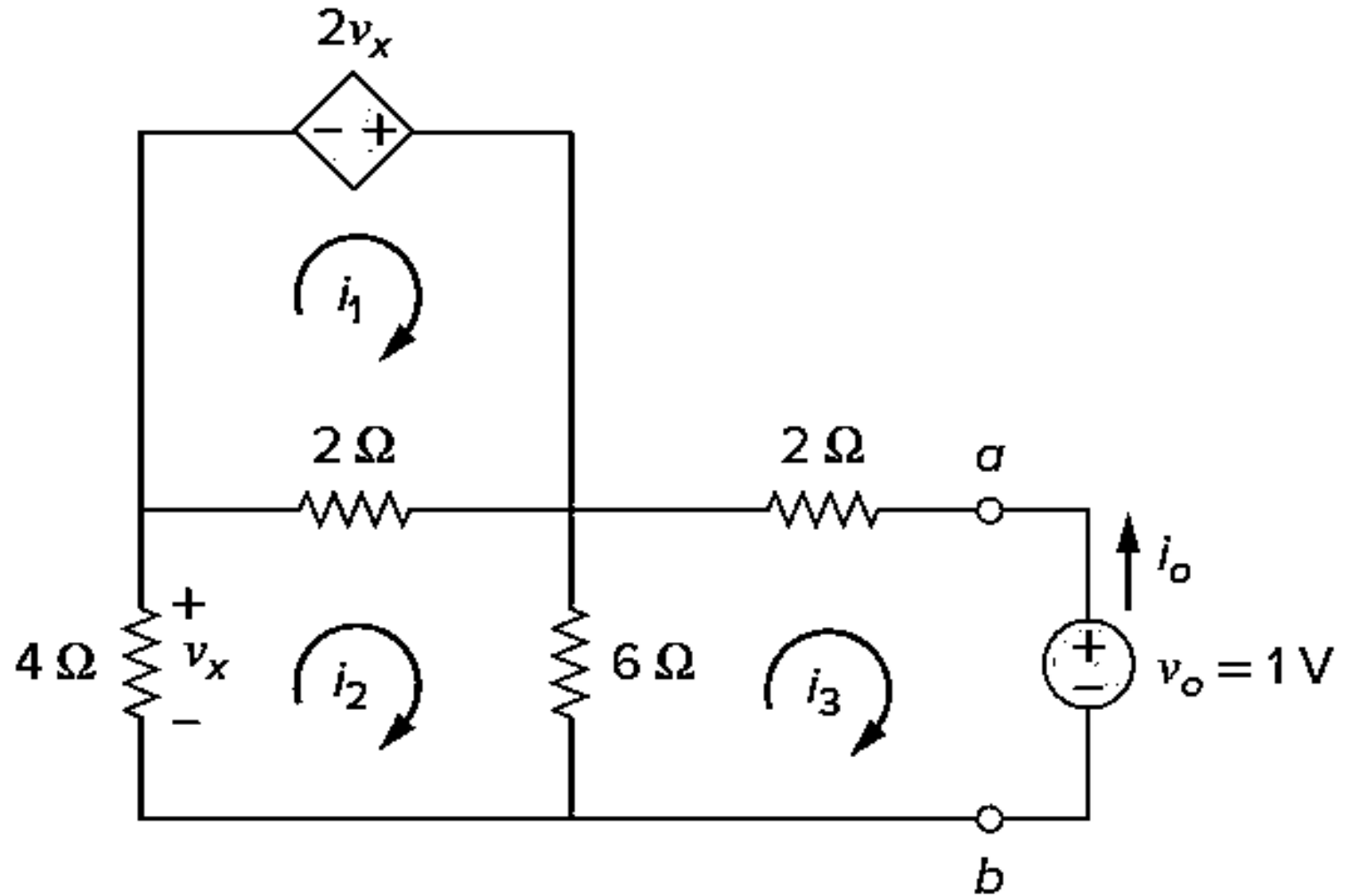
$$\Rightarrow v_x = i_1 - i_2$$

But,

$$-4i_2 = v_x = i_1 - i_2$$

Therefore,

$$i_1 = -3i_2$$



THEVENIN'S THEOREM

Mesh analysis for loop 2 and 3,

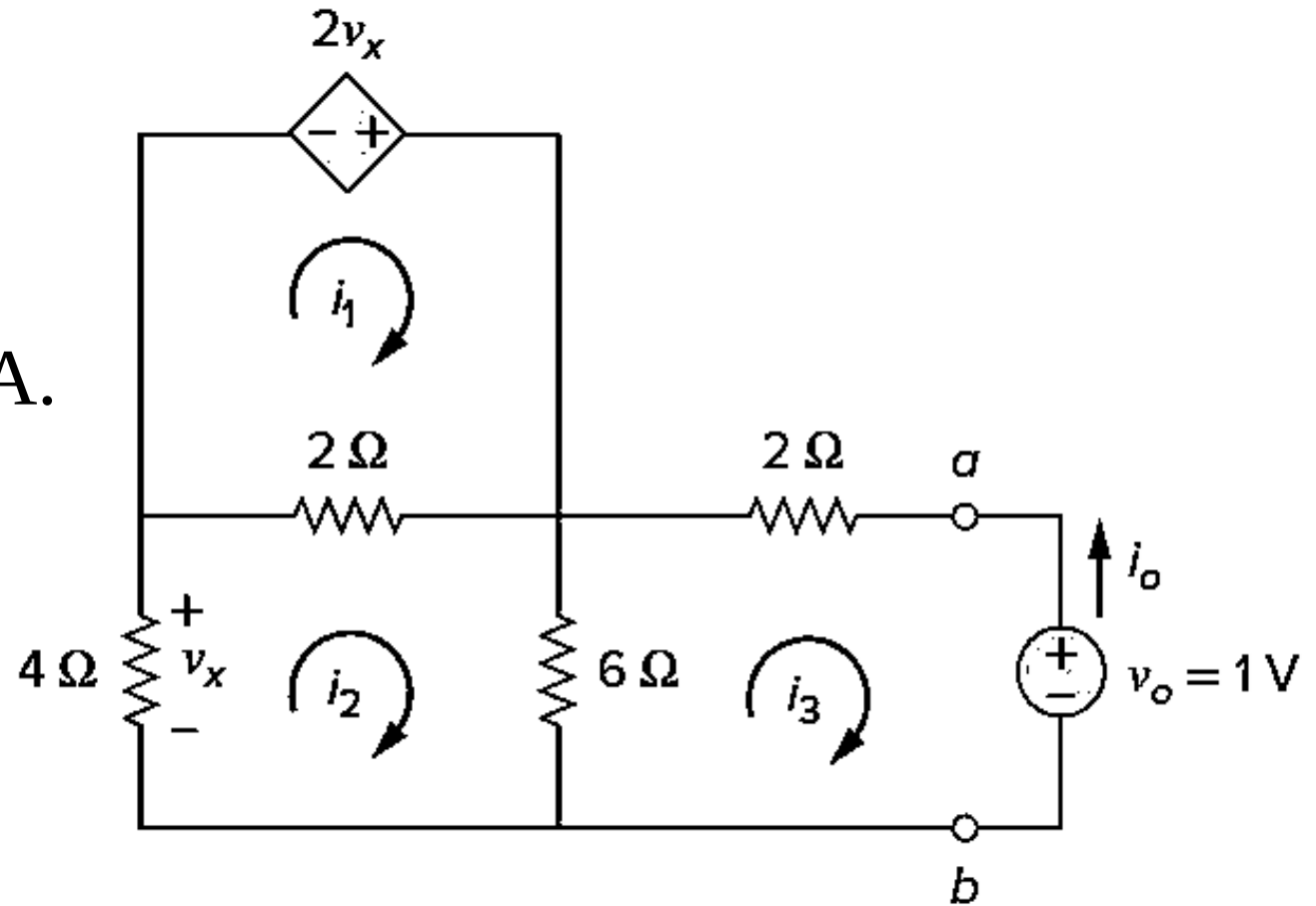
$$4i_2 + 2(i_2 - i_1) + 6(i_2 - i_3) = 0$$

$$\text{and } 6(i_3 - i_2) + 2i_3 + 1 = 0$$

On solving we get, $i_0 = -i_3 = 1/6$ A.

Therefore,

$$R_{Th} = \frac{v_o}{i_o} = \frac{1}{1/6} = 6\Omega$$



THEVENIN'S THEOREM

Step 3. Find V_{Th}

We have to find v_{oc} for this circuit.

Applying mesh analysis we get,

$$i_1 = 5 \quad (1)$$

$$-2v_x + 2(i_3 - i_2) = 0$$

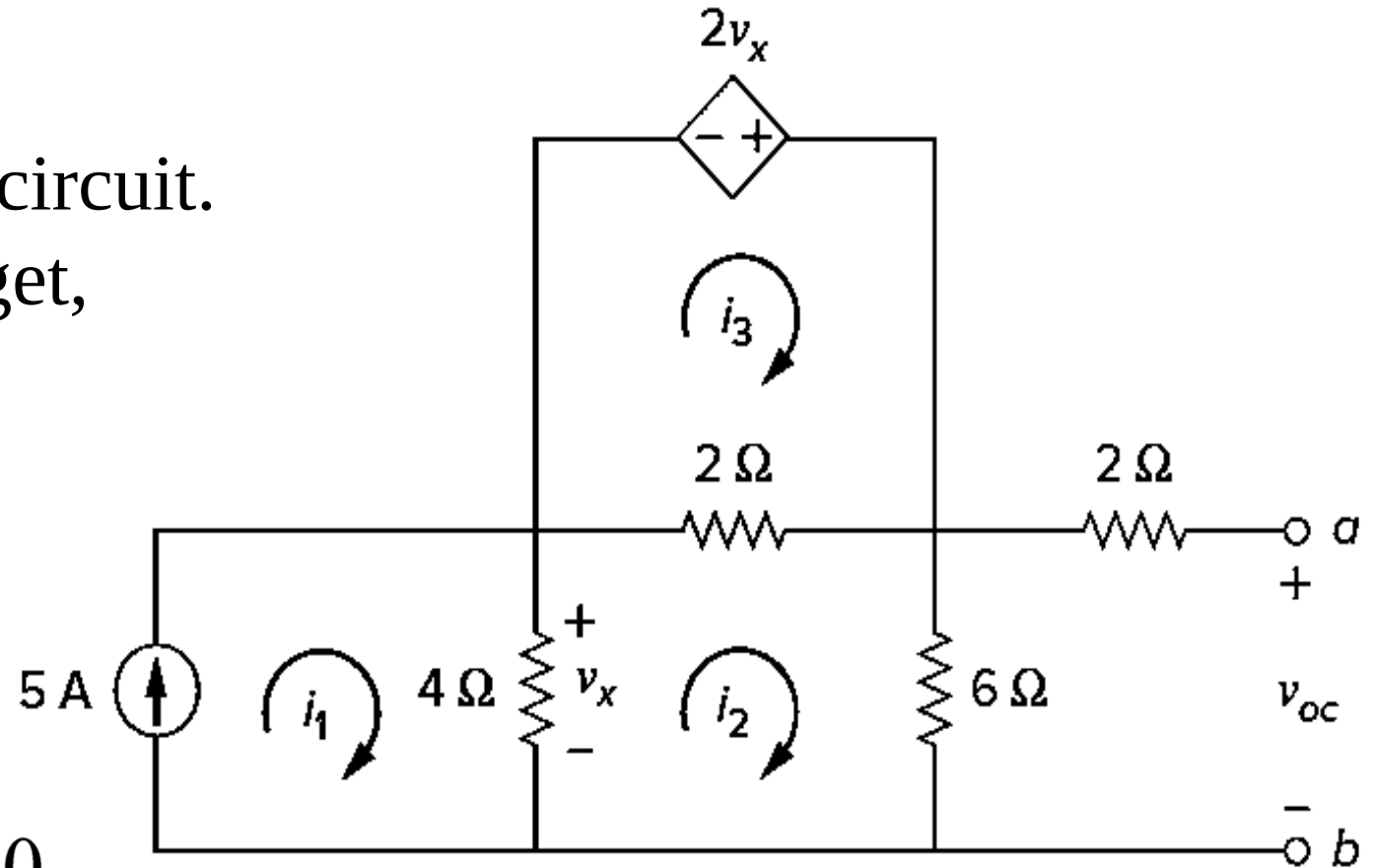
$$\Rightarrow v_x = i_3 - i_2 \quad (2)$$

and

$$4(i_2 - i_1) + 2(i_2 - i_3) + 6i_2 = 0$$

$$\Rightarrow 12i_2 - 4i_1 - 2i_3 = 0 \quad (3)$$

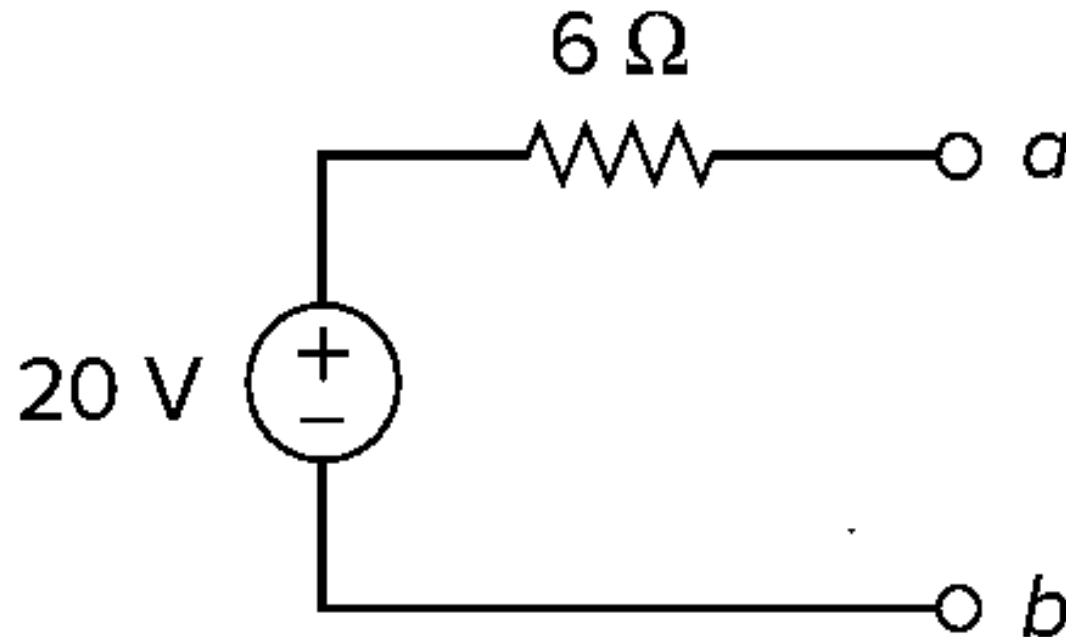
$$4(i_1 - i_2) = v_x \quad (4)$$



THEVENIN'S THEOREM

On solving (1),(2),(3) and (4), we get, $i_2 = 10/3 \Rightarrow V_{Th} = v_{oc} = 6i_2 = 20V$

The Thevenin equivalent circuit is shown below.



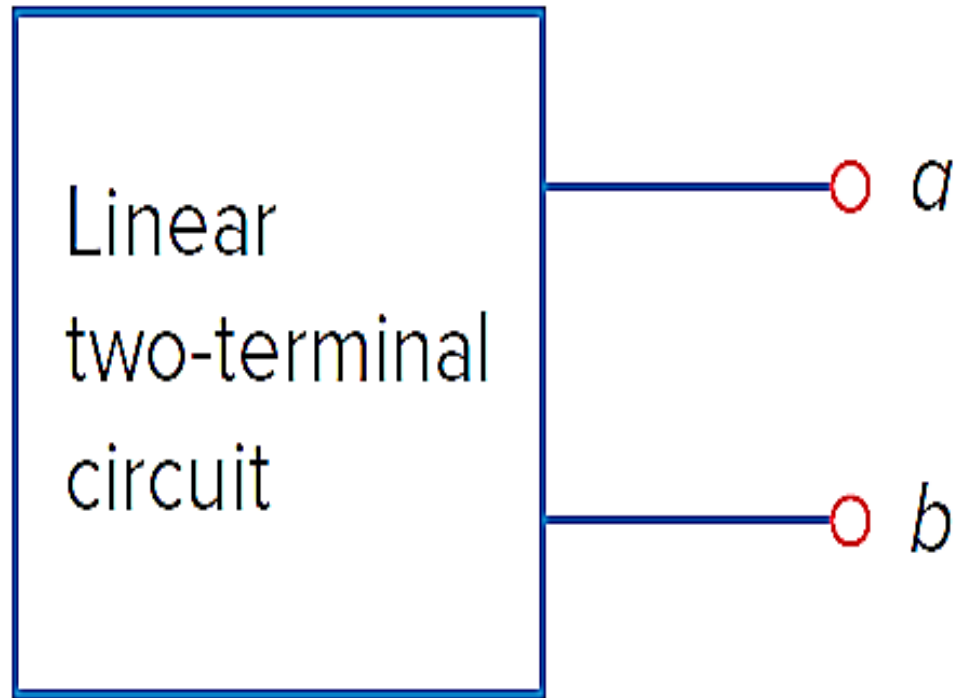
NORTON'S THEOREM

Statement: A linear two-terminal circuit can be replaced by an equivalent circuit consisting of a current source I_N in parallel with a resistor R_N , where I_N is the short-circuit current through the terminals and R_N is the input or equivalent resistance at the terminals when the independent sources are turned off.

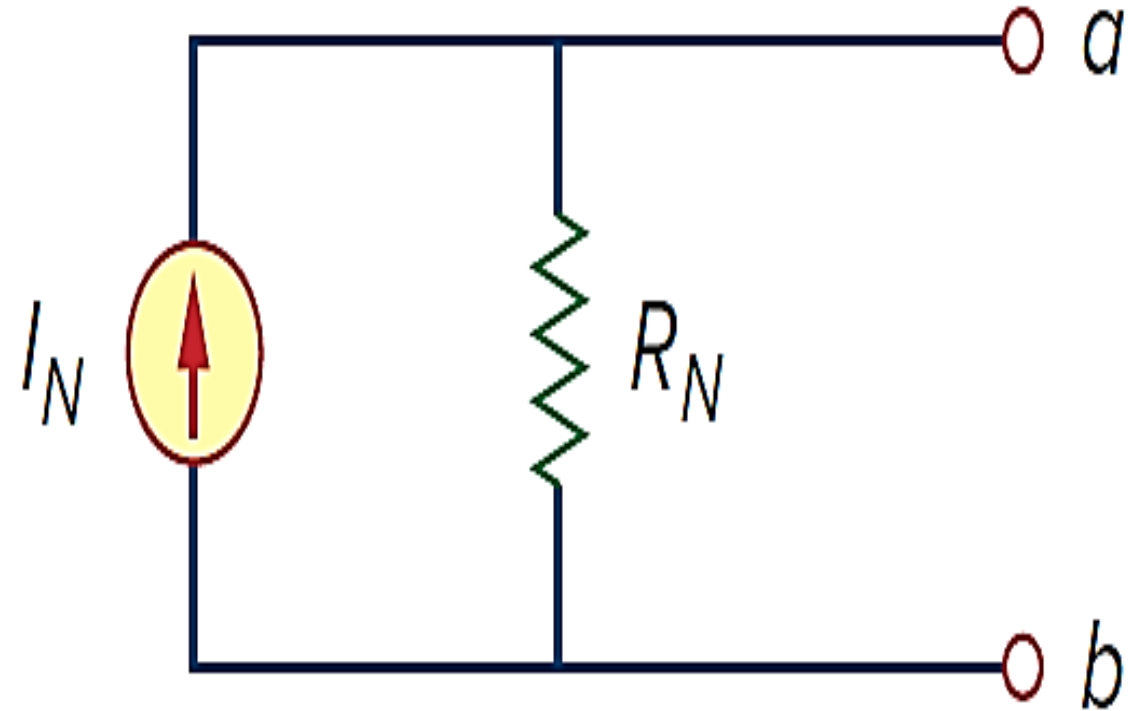
Why are we using Norton's Theorem?

- Simplifies the network in terms of currents instead of voltages.
- It reduces a network to a simple parallel circuit with a current source and a resistor.

NORTON'S THEOREM



Original circuit



Norton equivalent circuit

NORTON'S THEOREM

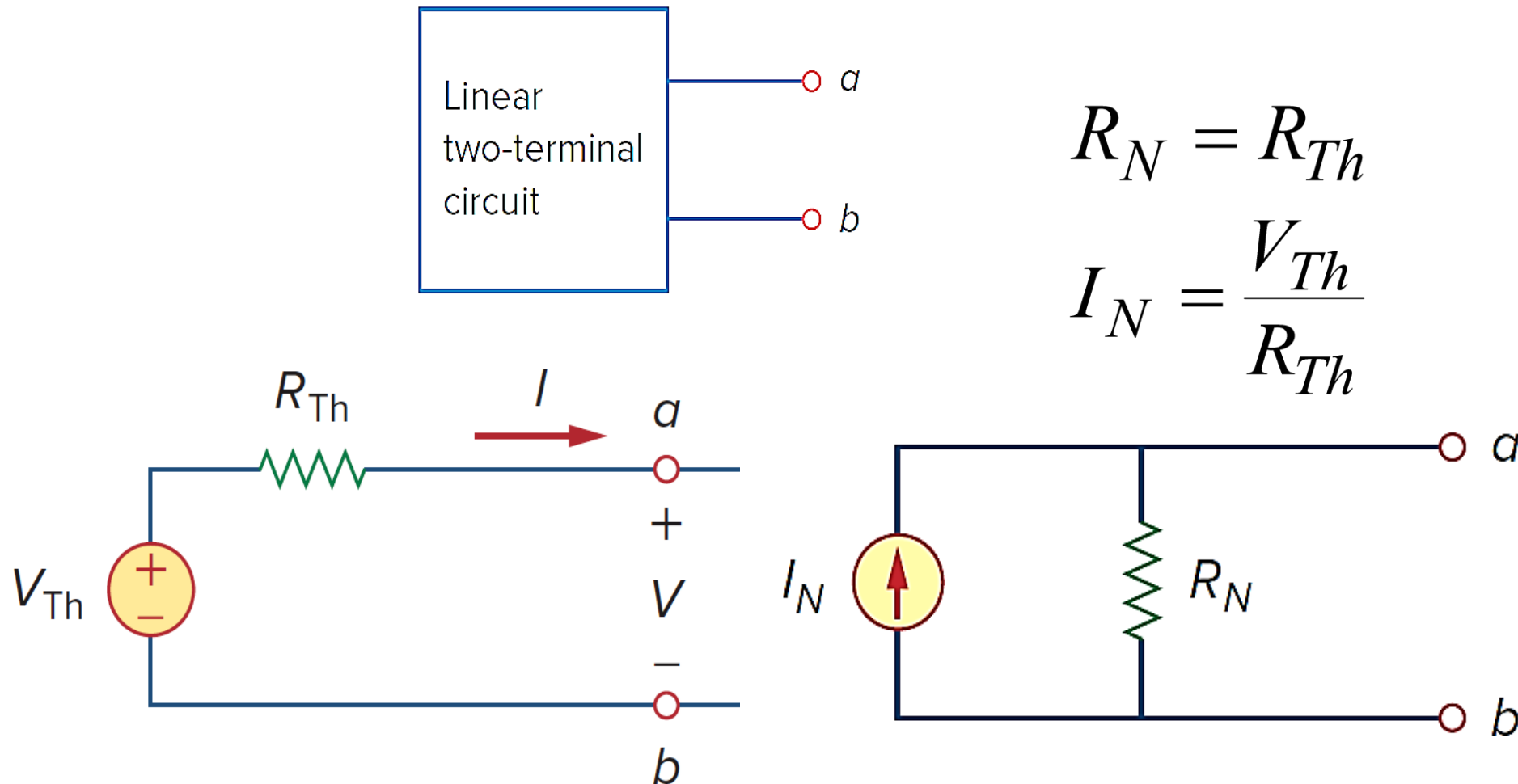
Steps to determine Norton's equivalent Resistance (R_N) and Current (I_N):

- Calculate R_N in the same way as R_{Th} .
- Using source transformation, the Thevenin and Norton resistances are equal i.e. $R_N = R_{Th}$.
- To find the Norton current I_N , we determine the short-circuit current flowing from terminal a to b .
- This short-circuit current is the Norton equivalent current I_N .

$$I_N = i_{SC}$$

NORTON'S THEOREM

Close relationship between Norton's and Thevenin's theorems:



NORTON'S THEOREM

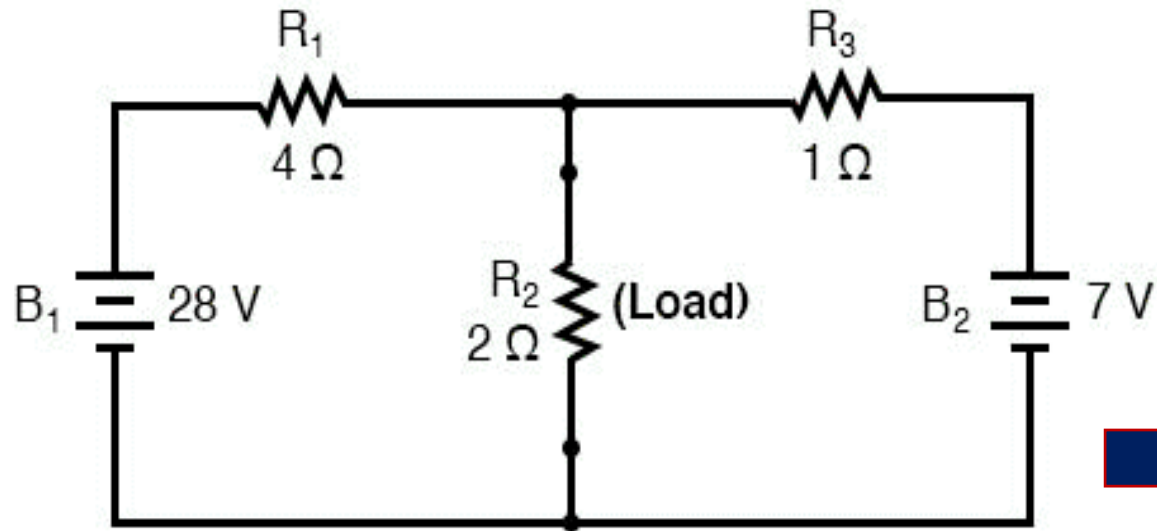
Since V_{Th} , I_N , and $R_{Th/N}$ are related, to determine the Thevenin or Norton equivalent circuit we find:

- The open-circuit voltage v_{oc} across terminals a and b ($= V_{Th}$)
- The short-circuit current i_{sc} at terminals a and b ($= I_N$)
- The equivalent input resistance R_{in} at terminals a and b when all independent sources are turned off ($= R_{Th/N}$)

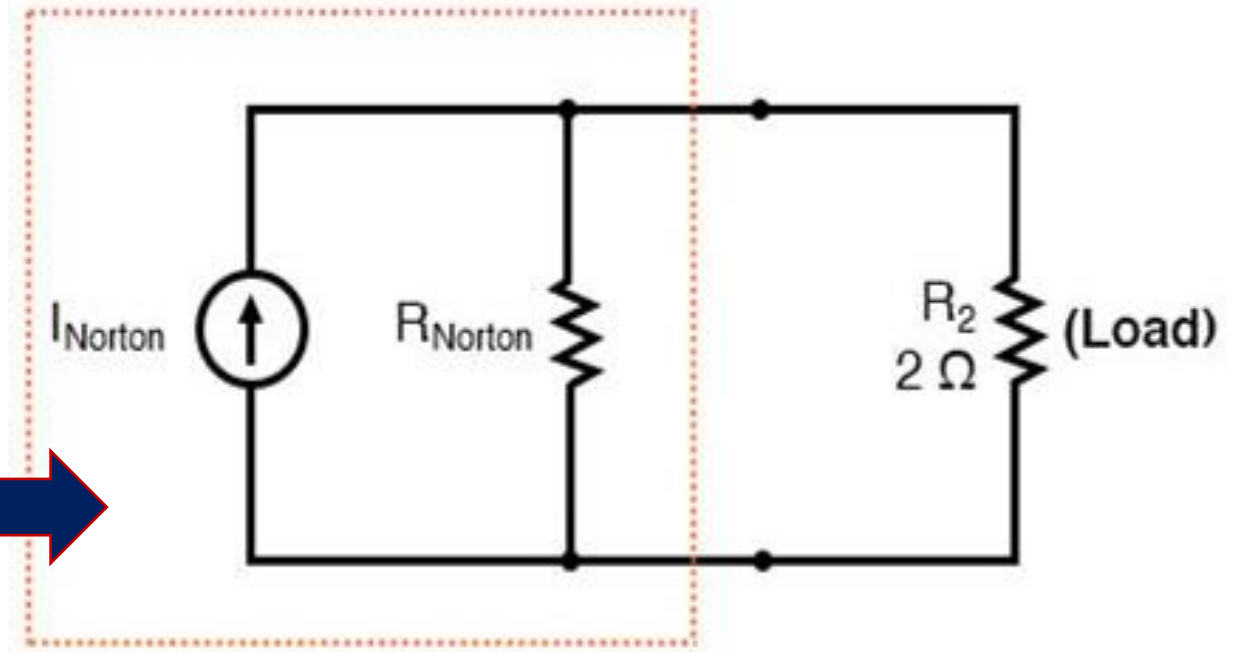
NORTON'S THEOREM

Example of Norton Theorem

Case 1: Without dependent source



Norton Equivalent Circuit

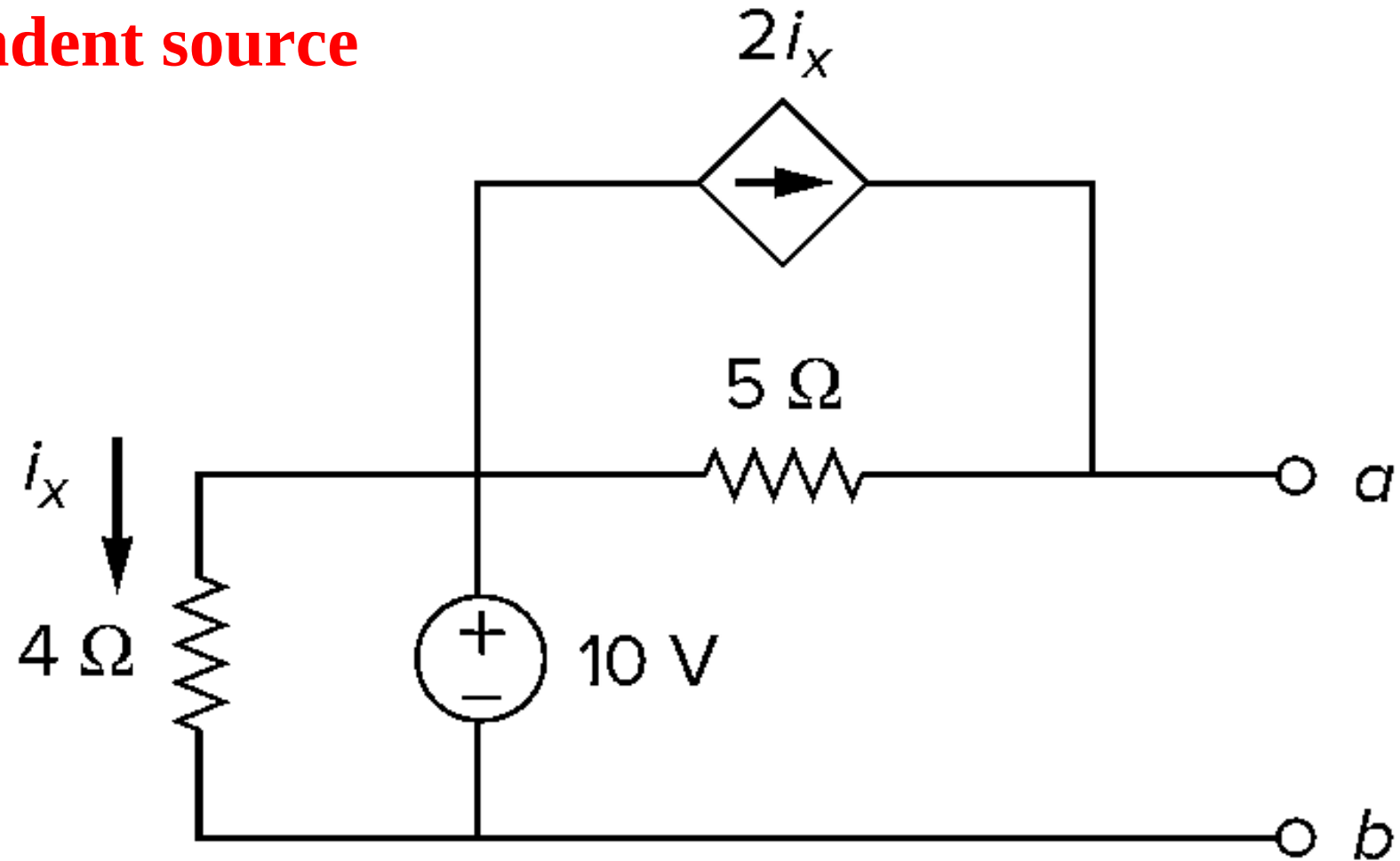


$$I_{\text{Norton}} = (28/4) + (7/1) = 14\text{ A}$$
$$R_{\text{Norton}} = (4||1) = 0.8\ \Omega$$

NORTON'S THEOREM

Example of Norton Theorem

Case 2: With dependent source

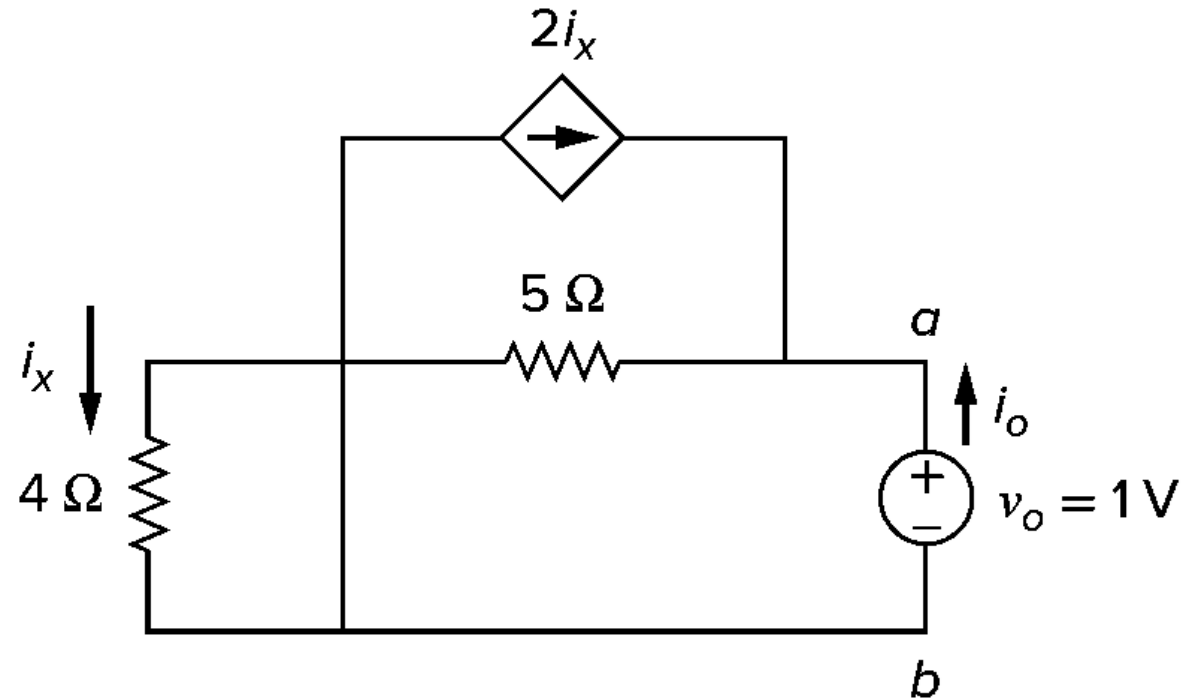


NORTON'S THEOREM

Step 1: Compute R_N . Set the independent sources equal to zero and connect a voltage source $v_o = 1V$ to the terminals.

We ignore the $4\text{-}\Omega$ resistor because it is short-circuited. Hence, $i_x = 0$.

Also due to the short circuit, the $5\text{-}\Omega$ resistor, the voltage source, and the dependent current source are all in parallel.



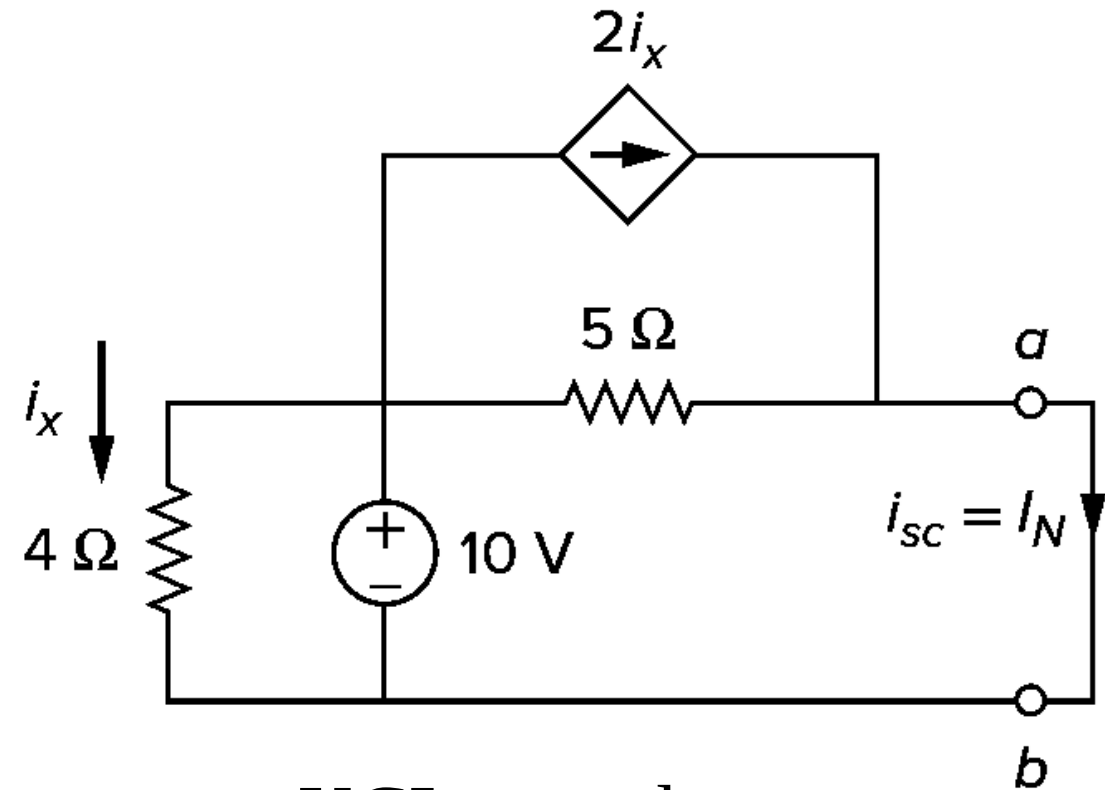
$$\text{At node a, } i_o = \frac{1}{5} = 0.2A$$

$$R_N = \frac{v_o}{i_o} = \frac{1}{0.2} = 5\Omega$$

NORTON'S THEOREM

Step 2: Compute I_N . Short-circuit terminals a and b and find the current i_{sc} , as indicated in the figure.

Note from this figure that the 4Ω resistor, the 10V voltage source, the 5Ω resistor, and the dependent current source are all in parallel.

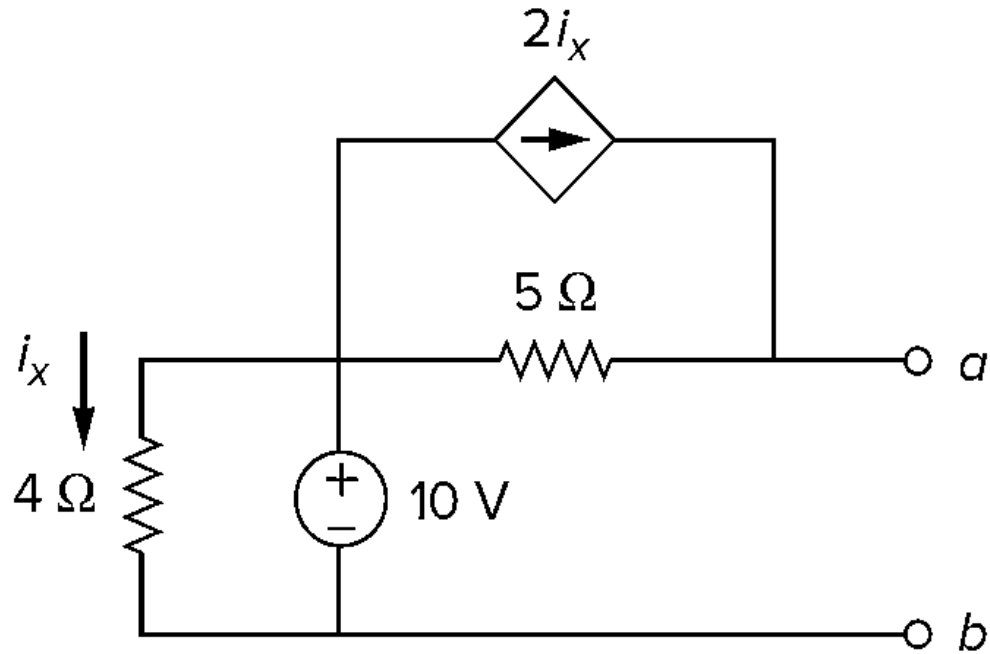


KCL at node a :

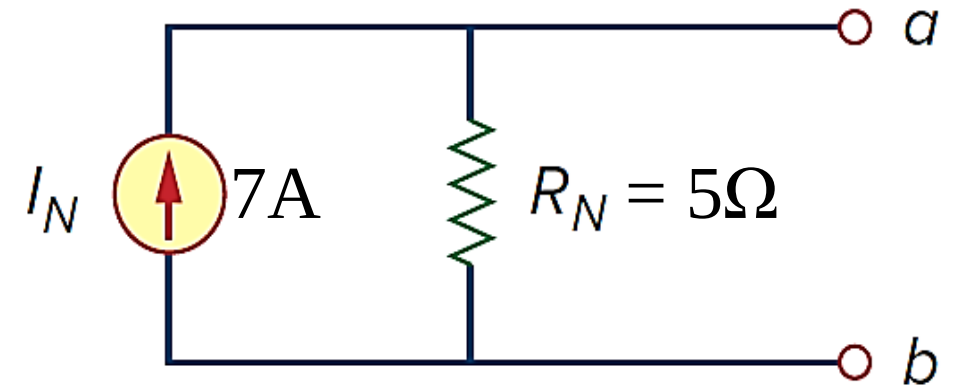
$$\text{Hence, } i_x = \frac{10}{4} = 2.5\text{A}$$

$$\boxed{I_N} = i_{sc} = \frac{10}{5} + 2i_x = 2 + 2 \times 2.5 = \boxed{7\text{A}}$$

NORTON'S THEOREM



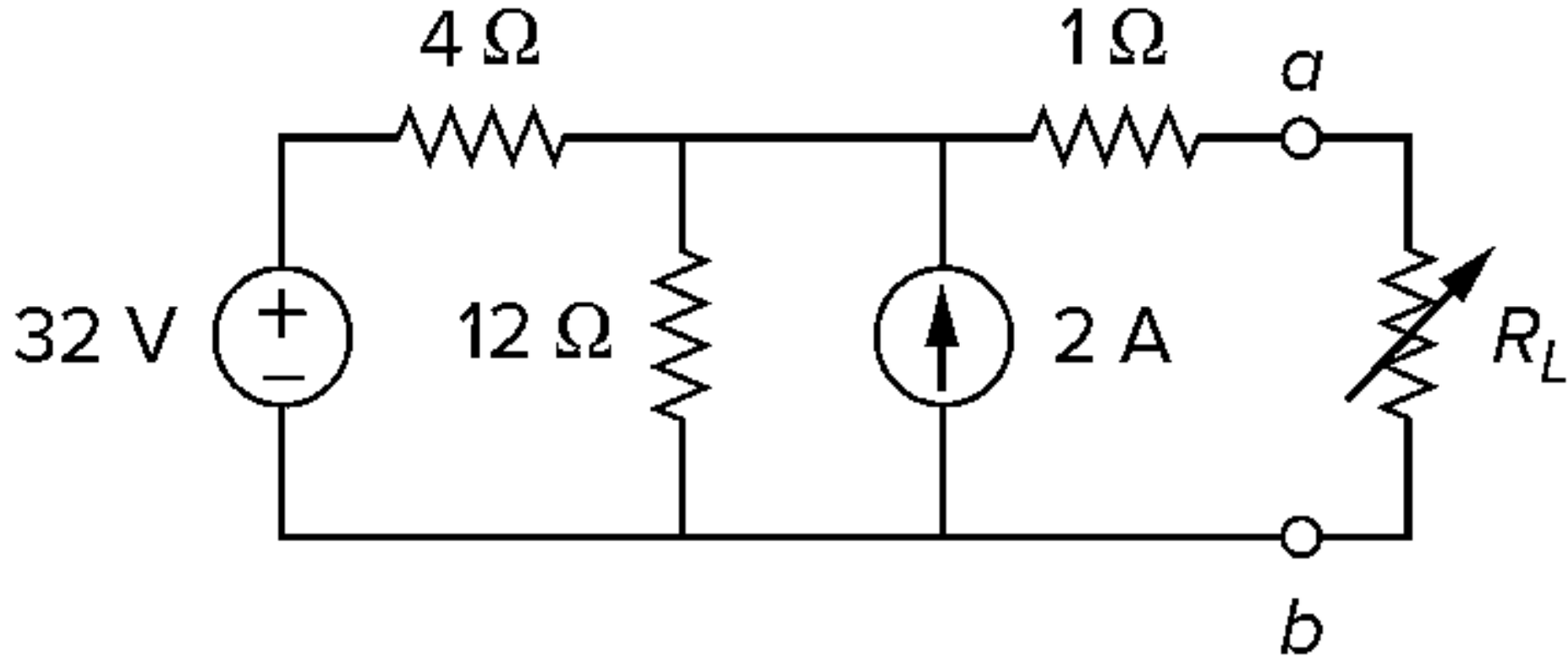
Original circuit



Norton equivalent

NUMERICAL

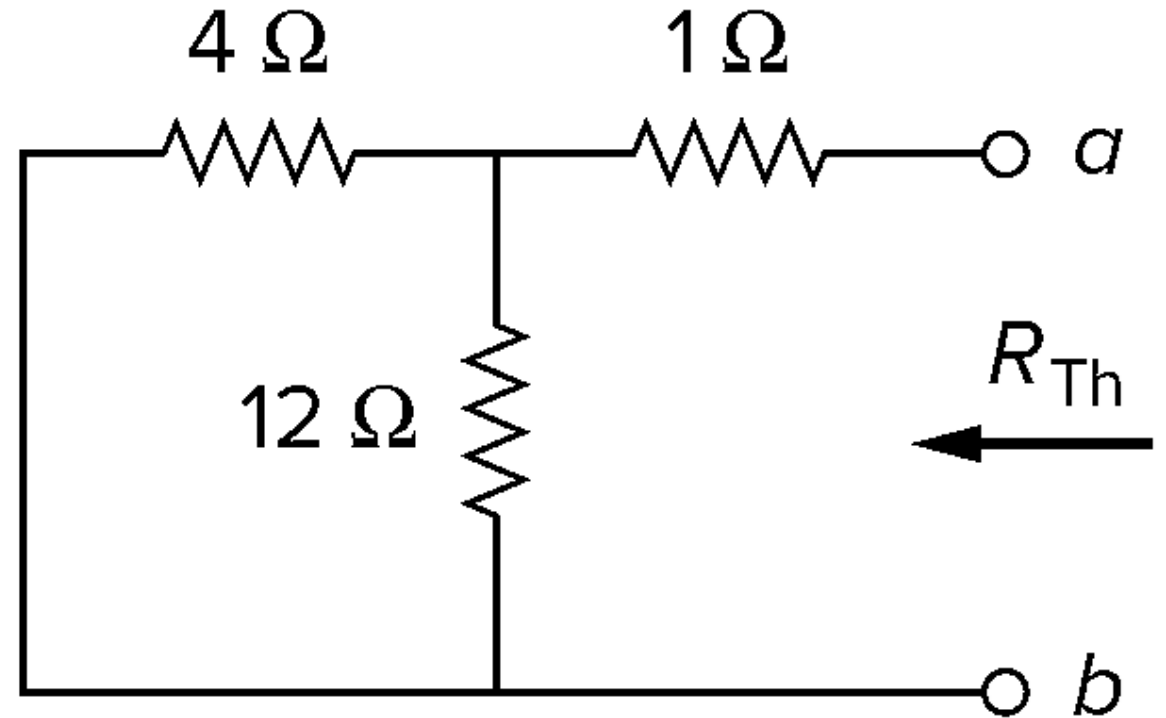
Q1. Find the Thevenin equivalent of the circuit shown below across terminals a - b . Then find the current through $R_L = 6\Omega$ and 36Ω respectively.



NUMERICAL

Soln:

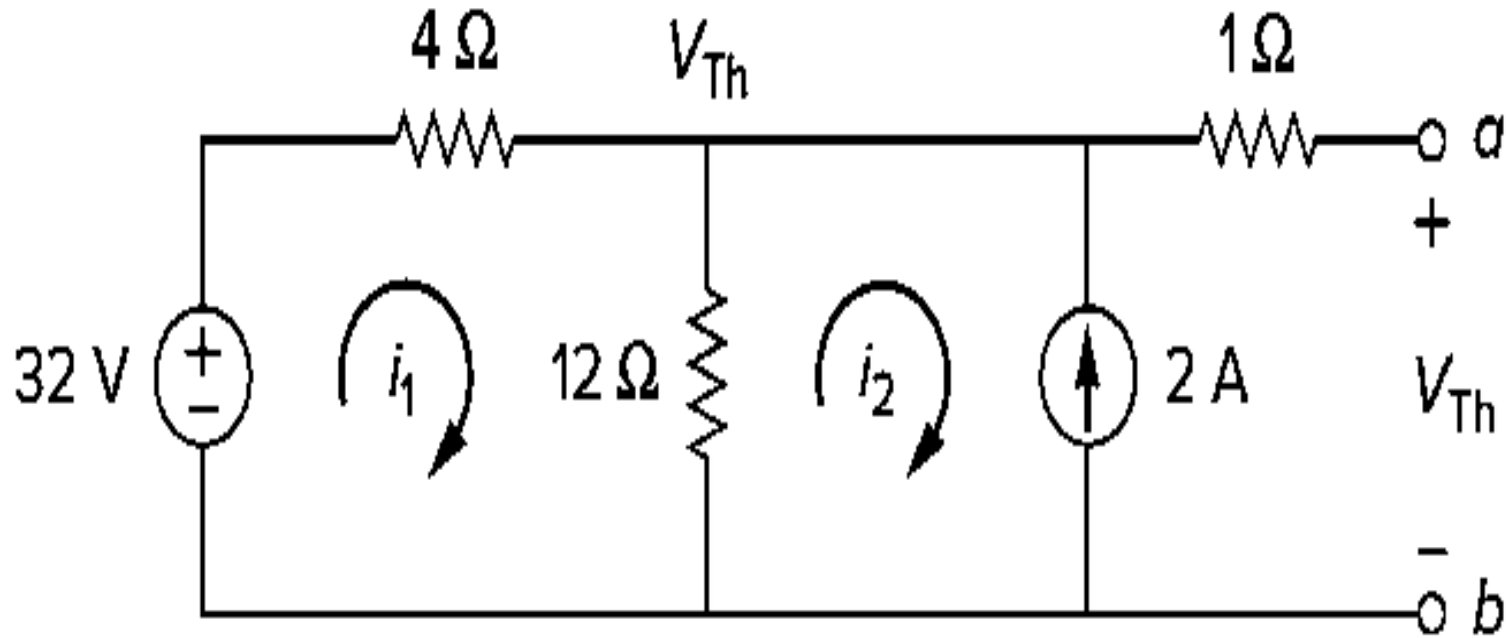
Step 1. Find R_{Th} by turning off the 32V voltage source (replacing it with a short circuit) and the 2A current source (replacing it with open circuit).



$$\boxed{R_{Th}} = 4 \parallel 12 + 1 = \frac{4 \times 12}{16} + 1 = \boxed{4\Omega}$$

NUMERICAL

Step 2. Make a - b open circuit. Find V_{Th} by applying mesh/node analysis.



$$-32 + 4i_1 + 12(i_1 - i_2) = 0, i_2 = -2A \Rightarrow i_1 = 0.5A$$

$$\boxed{V_{Th}} = 12(i_1 - i_2) = 12(0.5 + 2) = \boxed{30V}$$

NUMERICAL

Step 3. Finding current through R_L

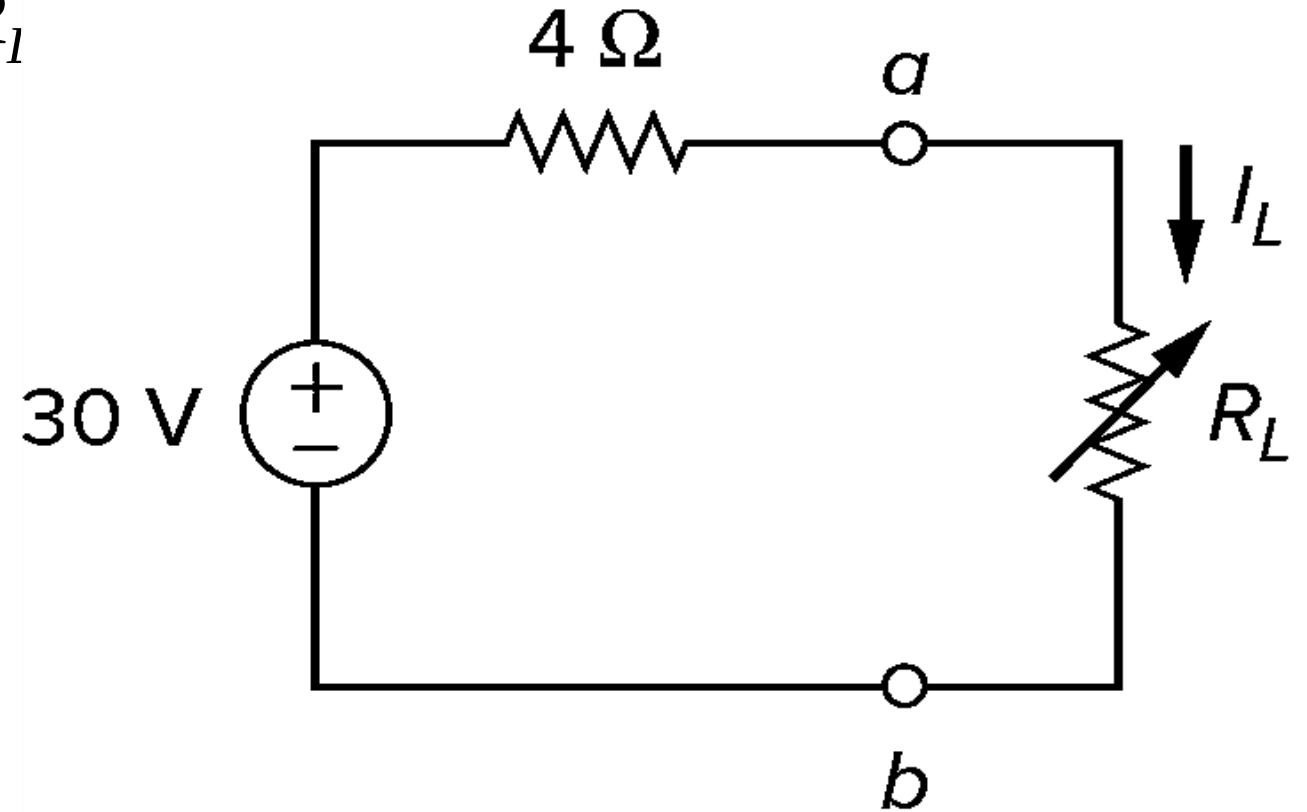
$$I_L = \frac{V_{Th}}{R_{Th} + R_L}$$

a) When $R_L = 6\Omega$

$$I_L = \frac{30}{4 + R_L} = \frac{30}{10} = 3A$$

a) When $R_L = 36\Omega$

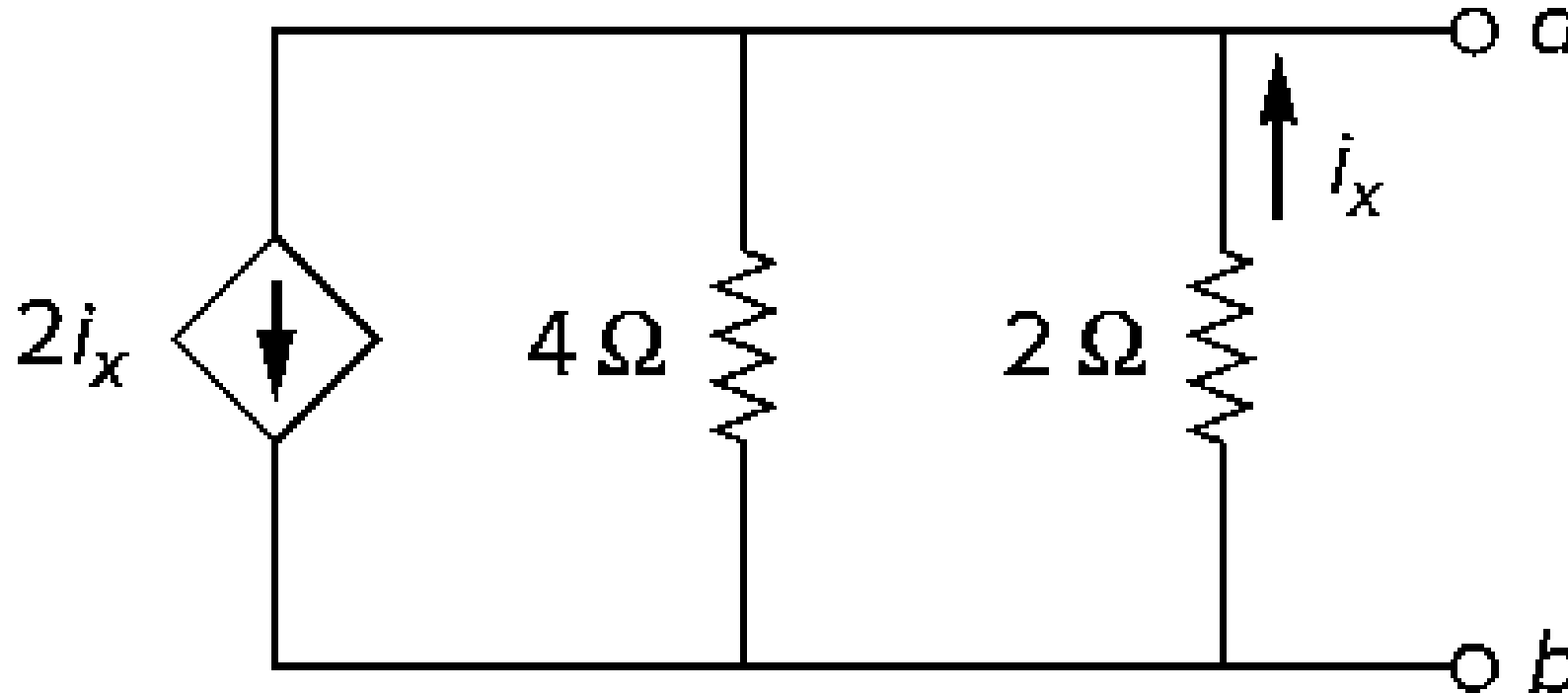
$$I_L = \frac{30}{4 + R_L} = \frac{30}{40} = 0.75A$$



Thevenin equivalent circuit

NUMERICAL

Q2. Find the Thevenin equivalent of the given circuit at terminals a - b .

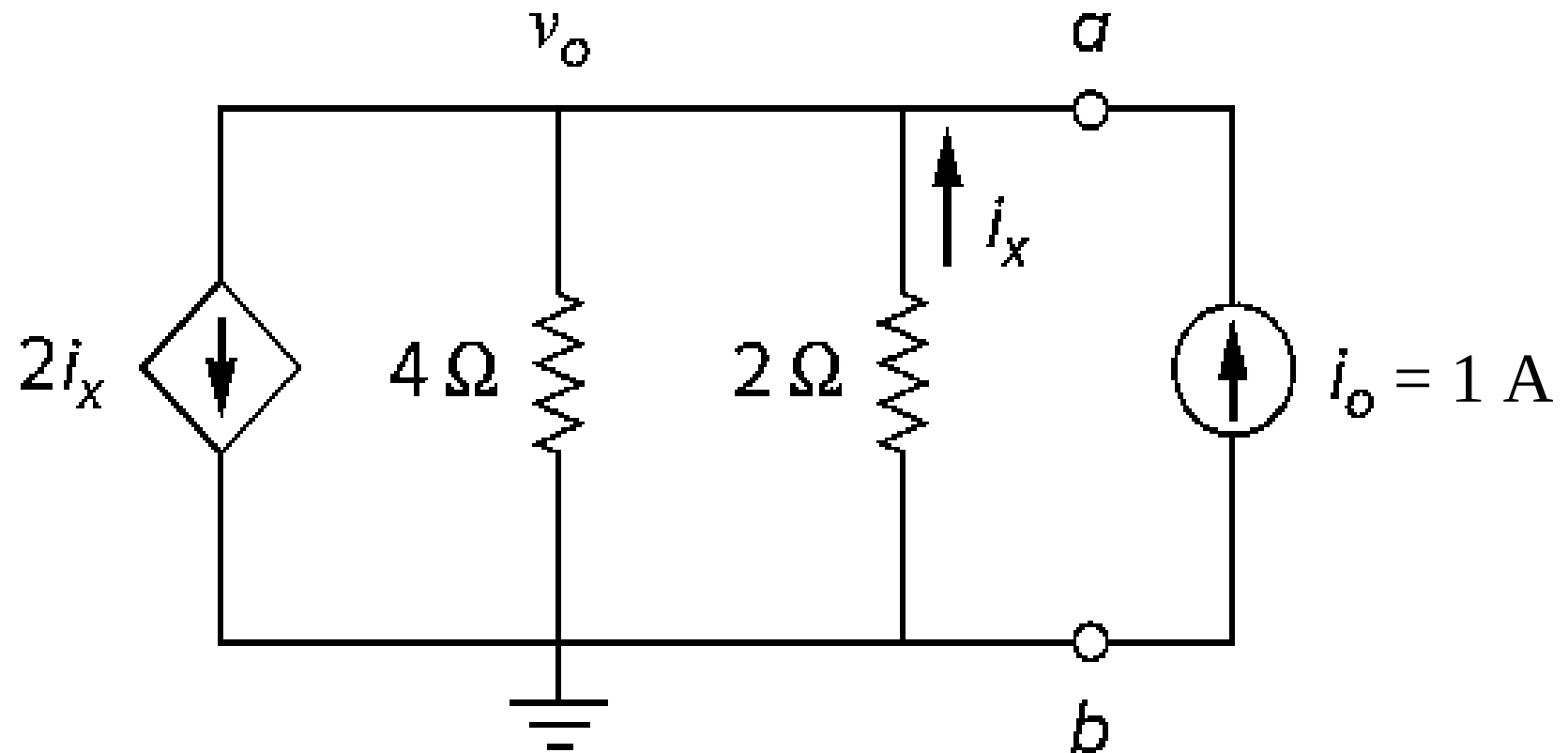


NUMERICAL

Soln:

Step 1. Find R_{Th} . We excite the circuit with 1A current source.

The reduced circuit is shown



NUMERICAL

KCL at node a :

$$2i_x + \frac{v_0 - 0}{4} + \frac{v_0 - 0}{2} + (-1) = 0 \quad (1)$$

Also we have,

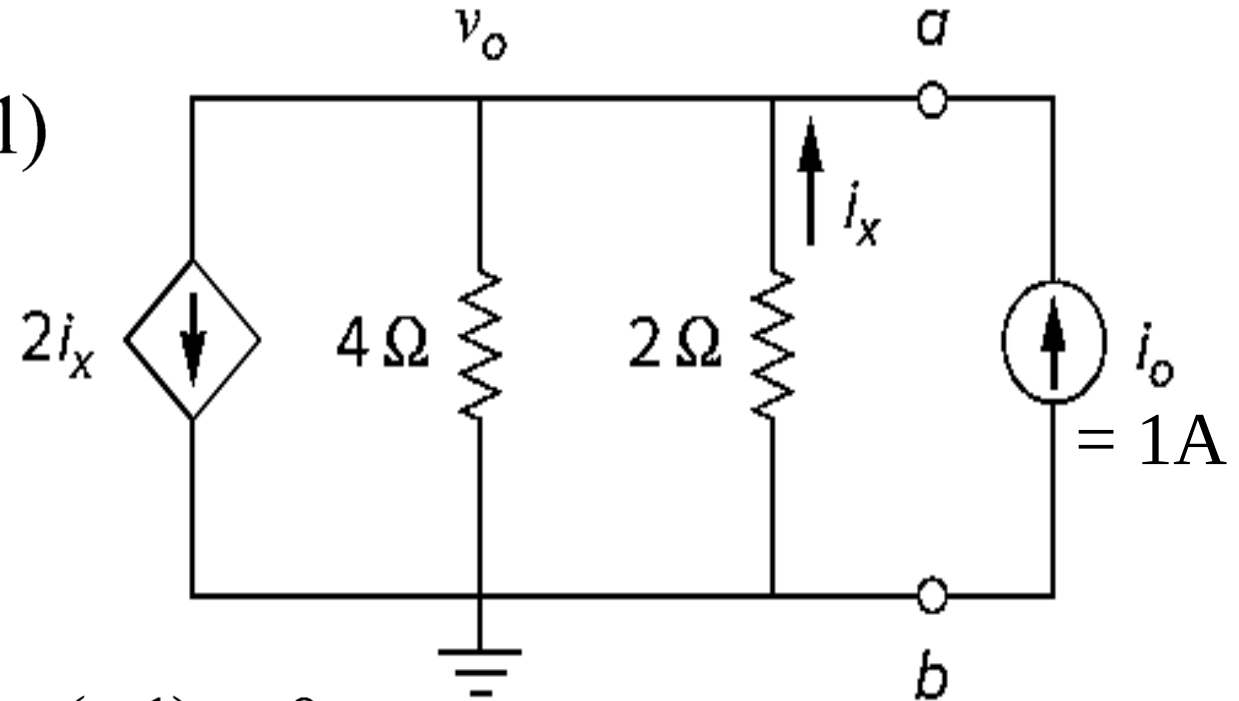
$$i_x = \frac{0 - v_0}{2} = -\frac{v_0}{2} \quad (2)$$

Substituting (2) in (1), we have

$$2(-v_0 / 2) + (v_0 - 0) / 4 + (v_0 - 0) / 2 + (-1) = 0$$

$$\Rightarrow v_0 = -4V$$

$$R_{Th} = \frac{v_0}{i_0} = \frac{(-4)}{1} = -4\Omega$$



Negative sign in resistance indicates that the circuit is supplying power. Resistors cannot supply power, it's the dependent current source that supplies power.

NUMERICAL

Step 2: Finding V_{Th}

KCL at node a :

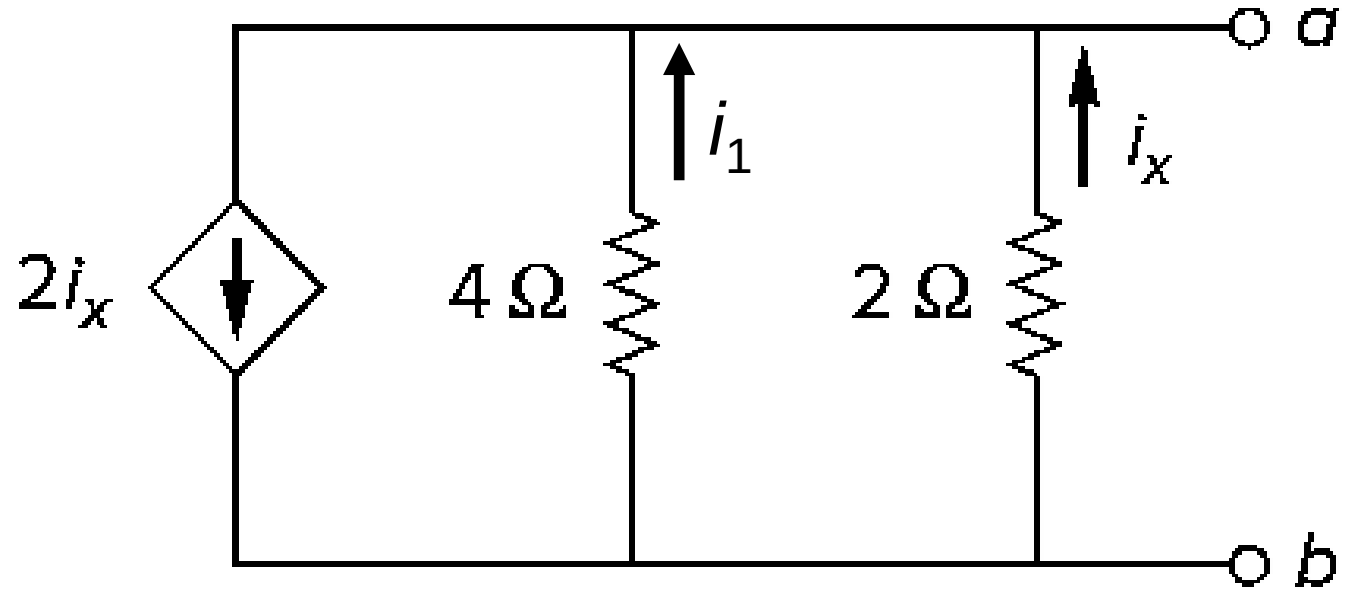
$$i_1 + i_x = 2i_x$$

$$\Rightarrow i_1 = i_x$$

$$\Rightarrow v_{ab} = -2i_x = -4i_x$$

$$\Rightarrow v_{ab} = i_x = 0$$

$$\Rightarrow \boxed{V_{Th}} = v_{ab} = \boxed{0}$$

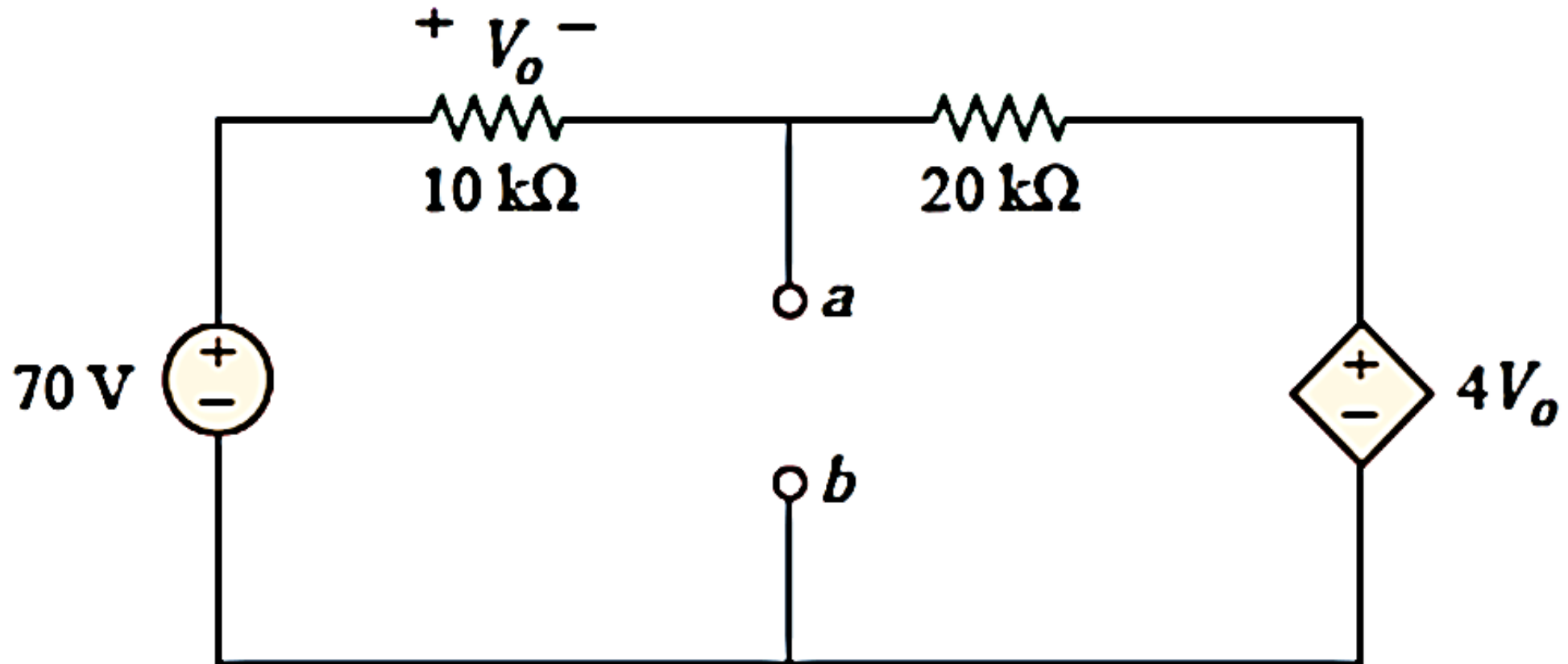


Since we don't have any independent sources in the given circuit, the value of Thevenin equivalent voltage V_{Th} **is zero**.

Thus, the Thevenin equivalent circuit is just a negative (-4Ω) resistance across $a-b$ i.e. a “dependent voltage source” with $v_{ab} = 4i_{ba}$

NUMERICAL

Q3. Find the Thevenin equivalent of the given circuit at terminals a - b .



NUMERICAL

Soln:

Step 1. Finding V_{Th} .

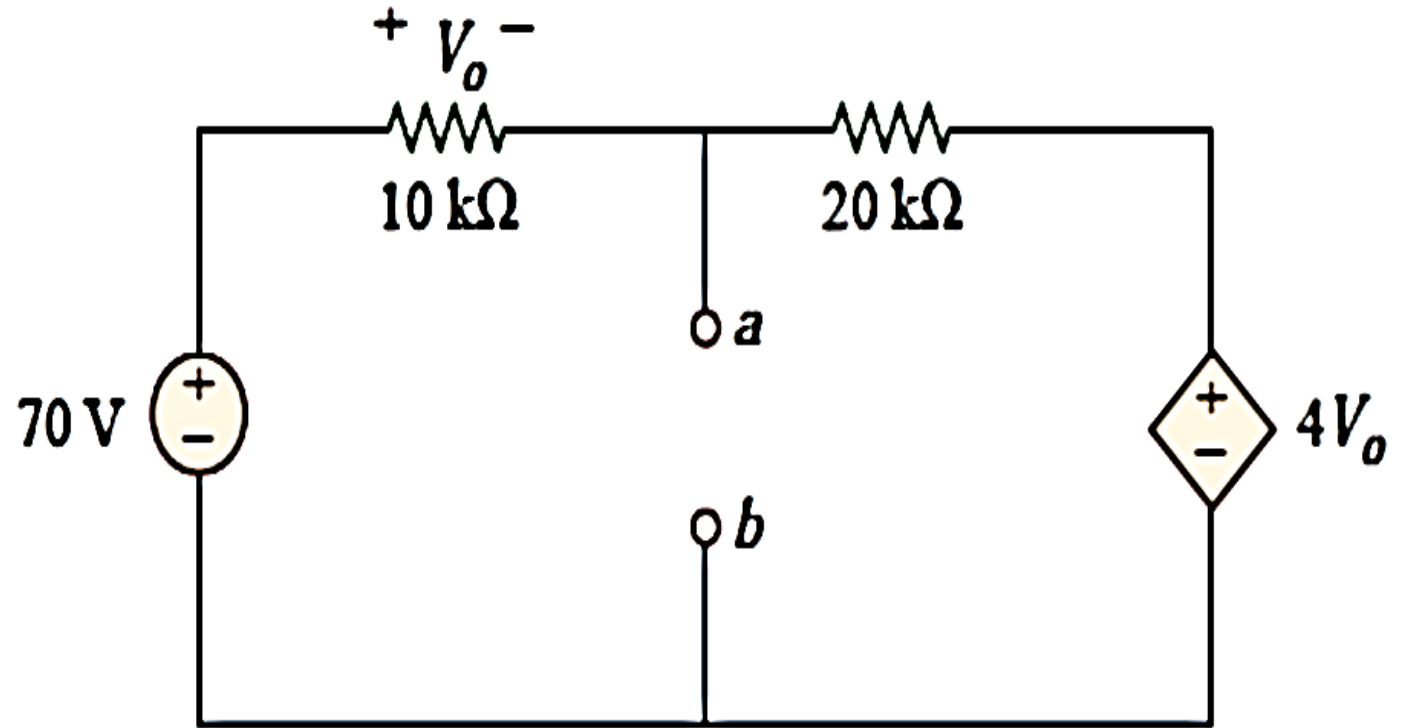
Applying KVL to the loop,

$$-70 + (10 + 20)kI + 4V_o = 0$$

But $V_o = 10k I$,

$$70 = 70kI \Rightarrow I = 1mA$$

$$-70 + 10kI + V_{Th} = 0 \Rightarrow V_{Th} = 60V$$



NUMERICAL

Step 2: To find R_{Th} , remove independent voltage source 70V and apply a 1-V independent source to excite the circuit at the terminals $a-b$

We have

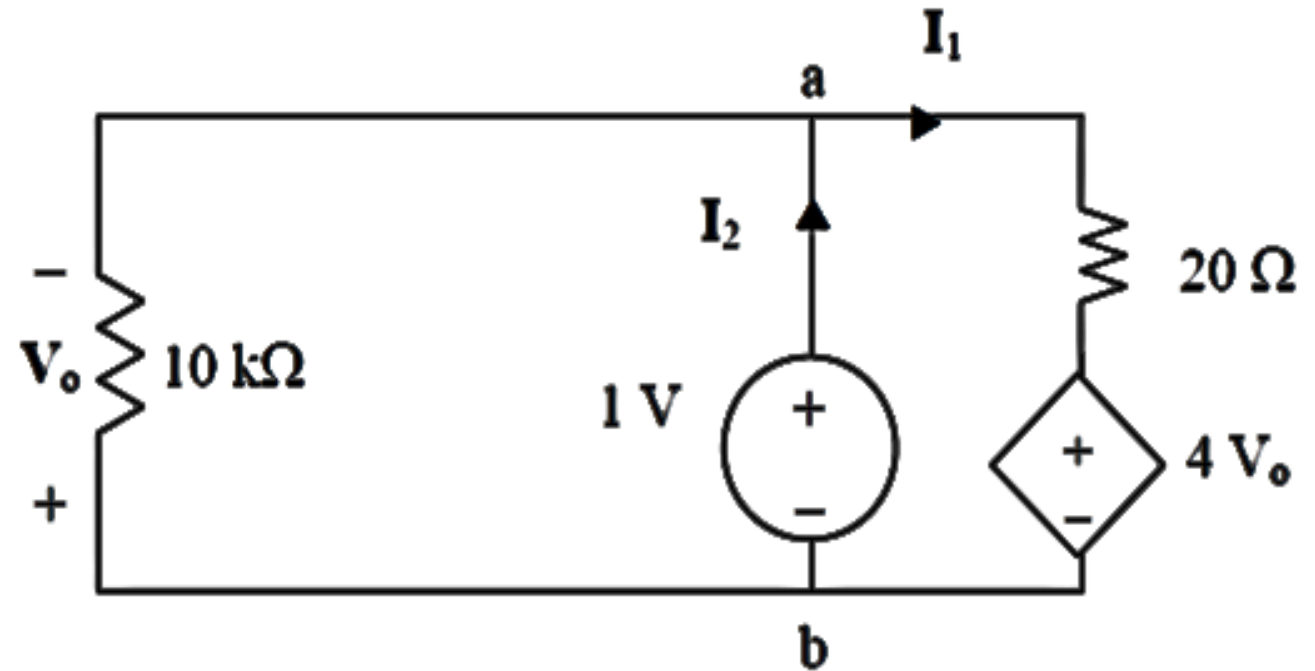
$$V_0 = -1V$$

$$-1 + 20kI_1 + 4V_0 = 0$$

$$\Rightarrow I_1 = 0.25mA$$

And

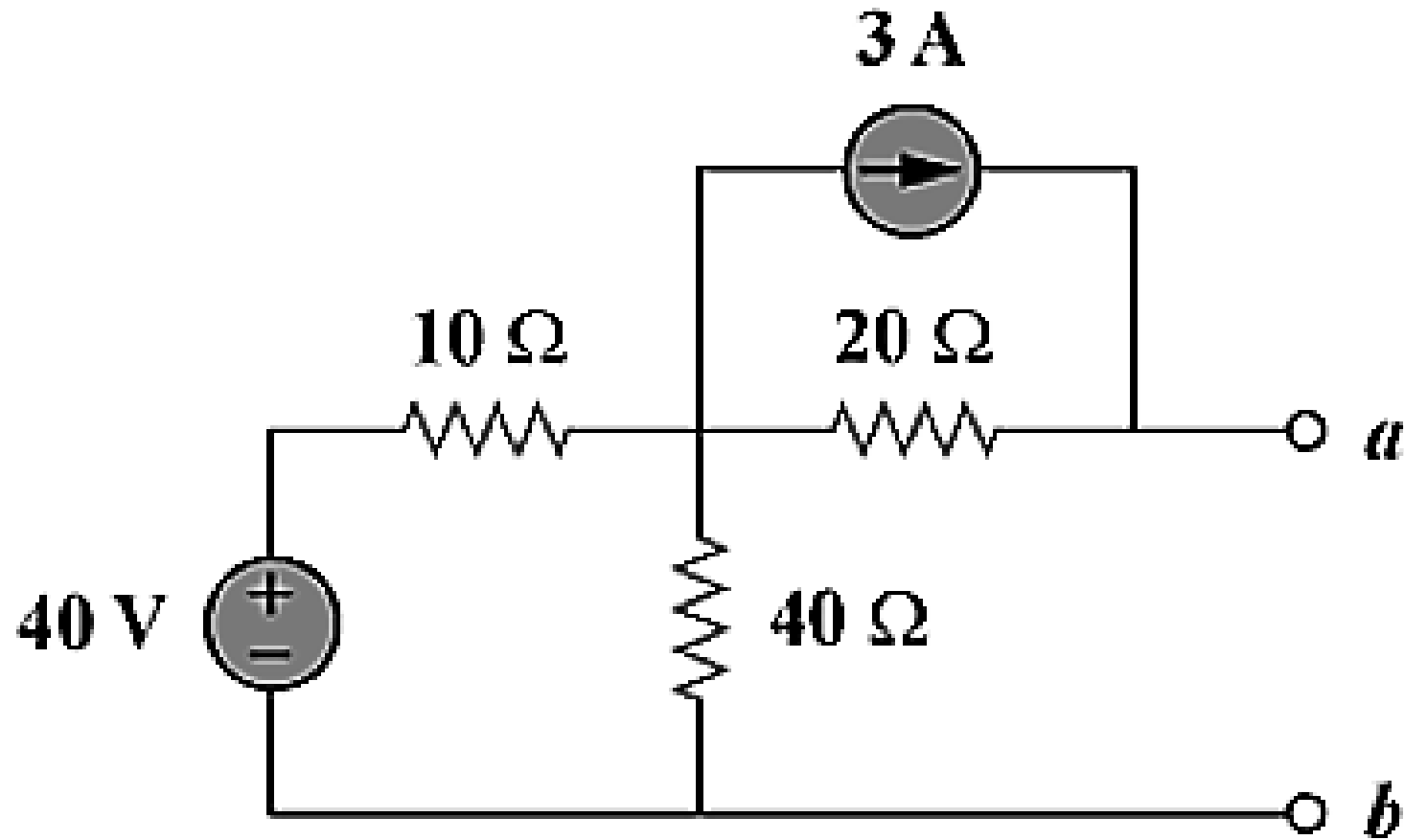
$$I_2 = I_1 + \frac{1}{10 \times 10^3} = 0.35mA$$



$$\boxed{R_{Th}} = \frac{1V}{I_2} = \frac{1}{0.35} \text{ k}\Omega = \boxed{2.857 \text{ k}\Omega}$$

NUMERICAL

Q4. Determine the Thevenin equivalent circuit between terminals a - b .



NUMERICAL

Soln:

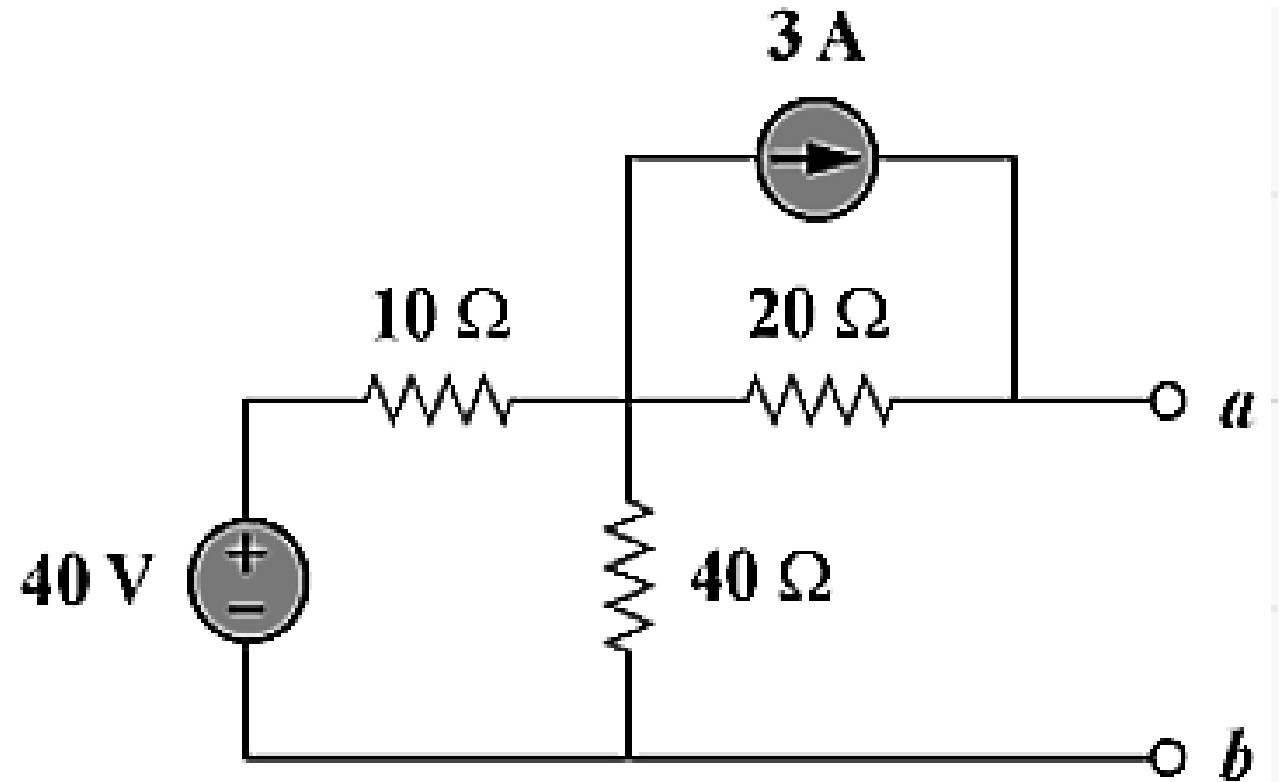
Step 1: Finding R_{Th} .

Remove independent sources
i.e replace voltage source by
short circuit and current
source by open circuit.

Then we have,

$$R_{Th} = 20 + 10 \parallel 40 = 20 + 400 / 50$$

$$\Rightarrow R_{Th} = 28\Omega$$



NUMERICAL

Step 2: Finding V_{Th} .

KCL at node 1:

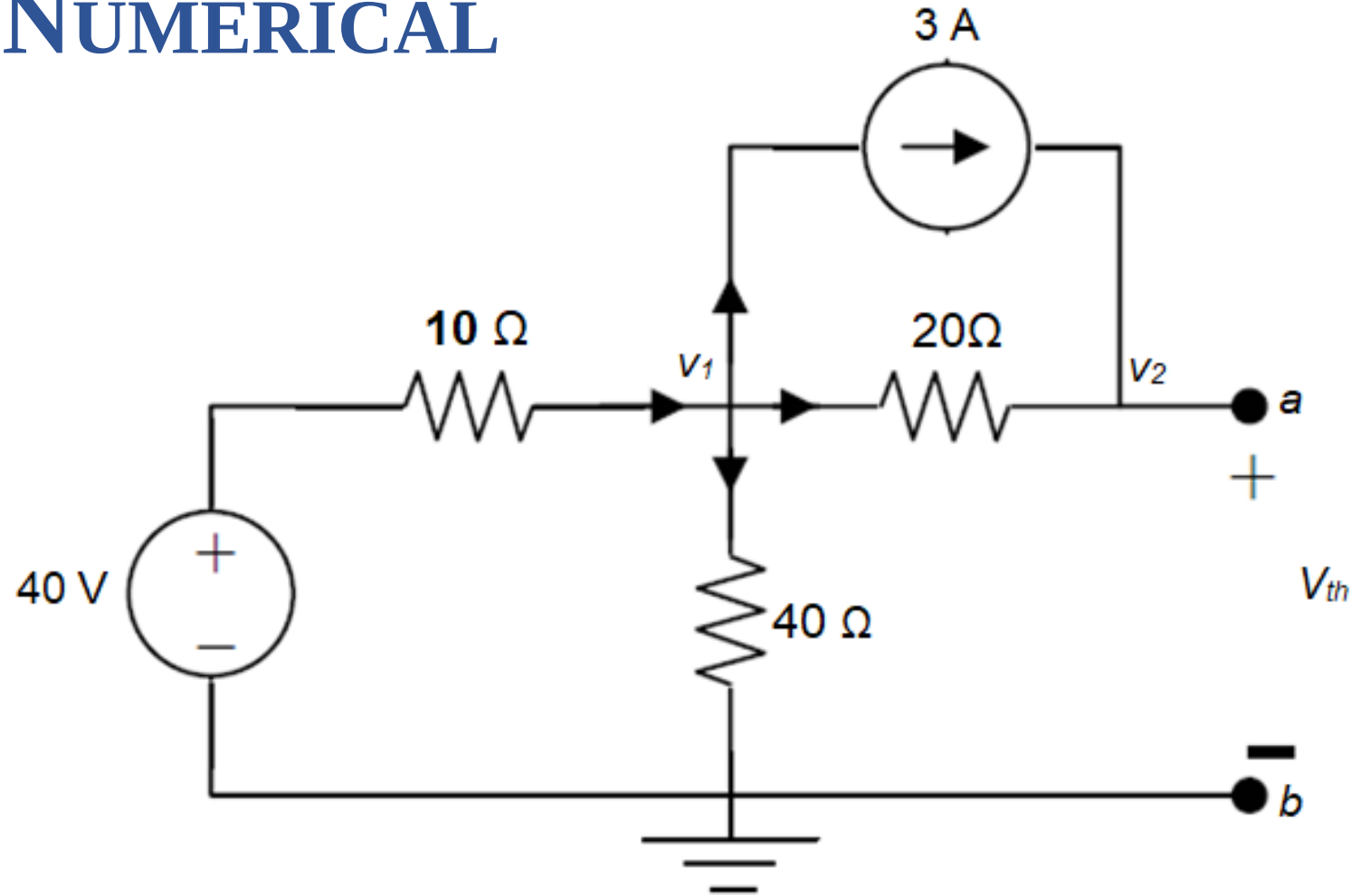
$$\frac{40 - v_1}{10} = 3 + \frac{v_1 - v_2}{20} + \frac{v_1}{40}$$
$$\Rightarrow 40 = 7v_1 - 2v_2 \quad (1)$$

KCL at node 2:

$$3 + \frac{v_1 - v_2}{20} = 0$$
$$\Rightarrow v_1 = v_2 - 60 \quad (2)$$

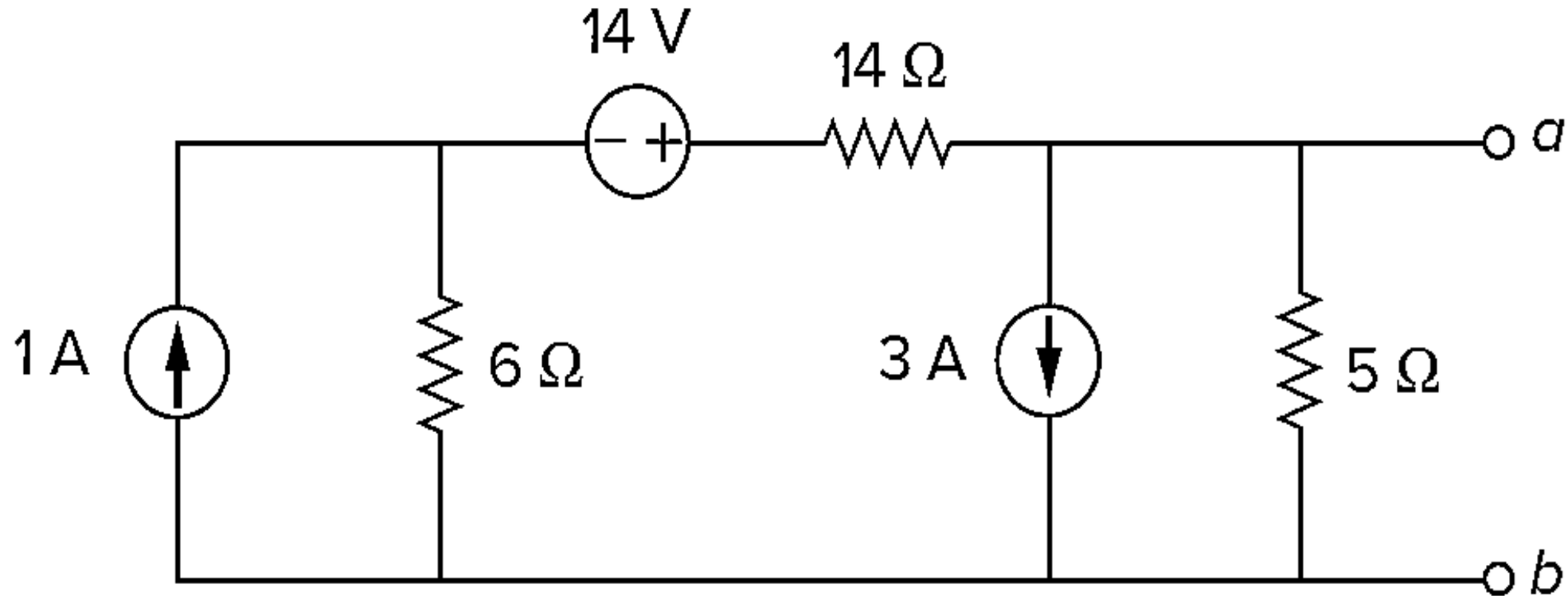
Solving (1) and (2), $v_1 = 32\text{V}$
 $v_2 = 92\text{V}$

$$\Rightarrow V_{Th} = v_2 = 92\text{V}$$



NUMERICAL

Q5. Find the Thevenin and Norton equivalents at terminals a - b of the circuit shown below.



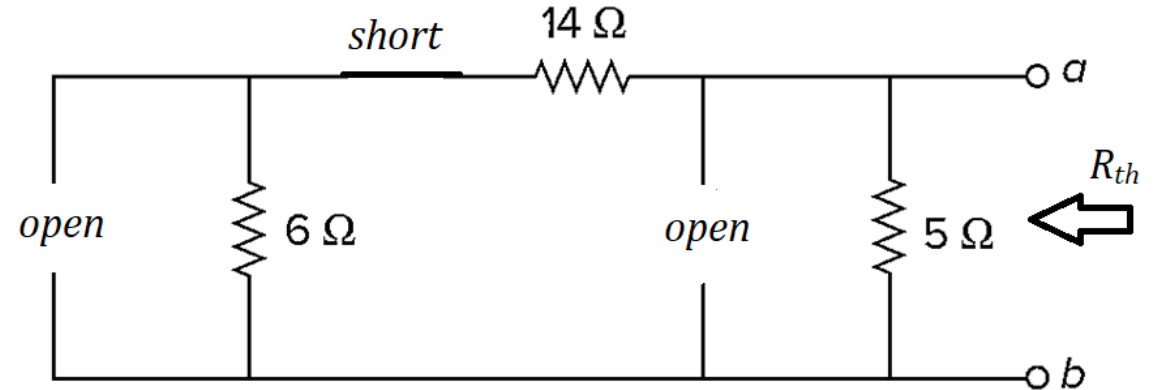
NUMERICAL

Soln:

Step 1: Finding $R_{Th/N}$. Replace current sources by open-circuits and voltage source by short-circuit.

$$R_{Th} = 5 \parallel (14 + 6) = \frac{20 \times 5}{20 + 5}$$

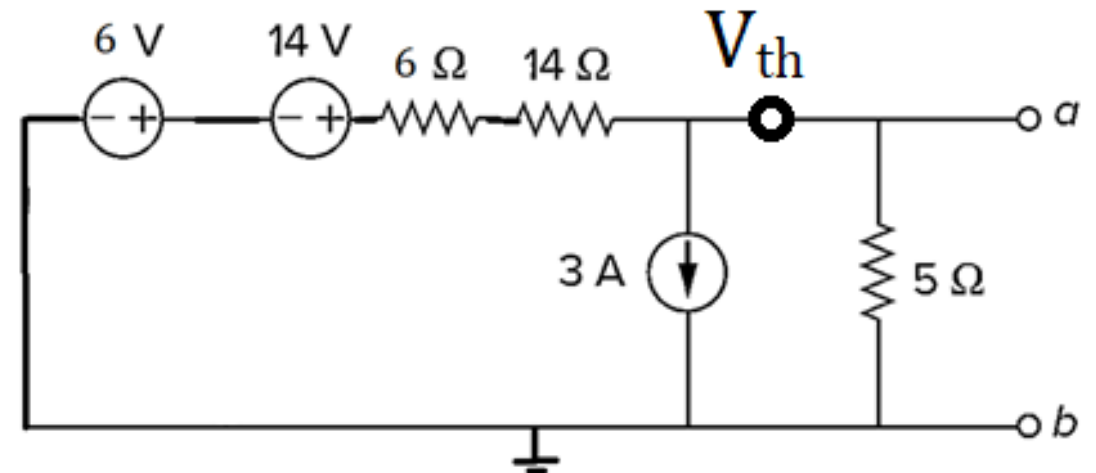
$$\Rightarrow R_{Th} = 4 \Omega = R_N$$



Step 2: Finding V_{Th} . Apply source transformation to 1A current source and apply nodal analysis.

$$\frac{14 + 6 - V_{Th}}{14 + 6} = 3 + \frac{V_{Th}}{5}$$

$$\Rightarrow V_{Th} = -8V$$



NUMERICAL

And,

$$I_N = \frac{V_{Th}}{R_{Th}} = \frac{-8}{4}$$

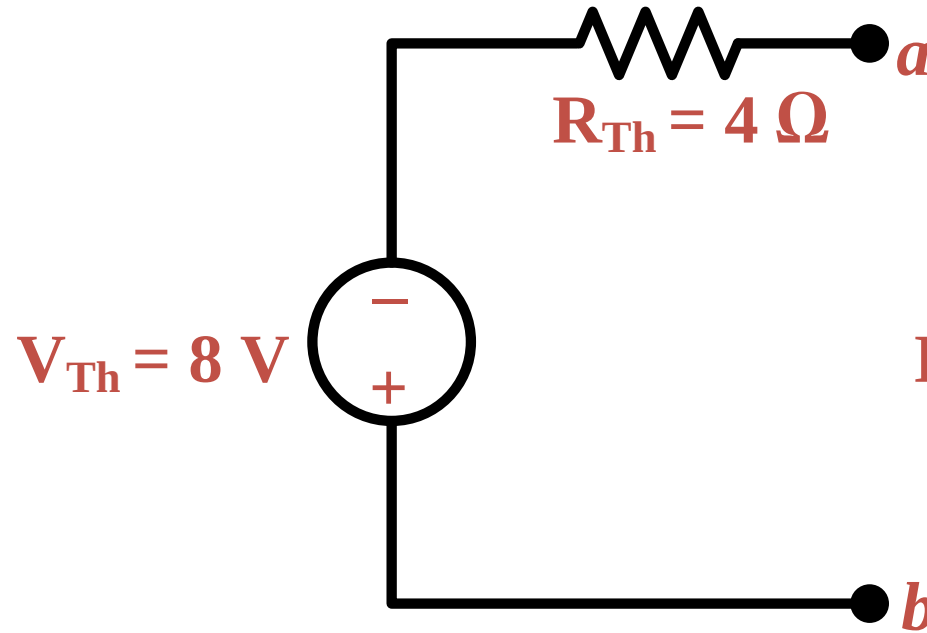
$$\Rightarrow I_N = -2A$$

Therefore,

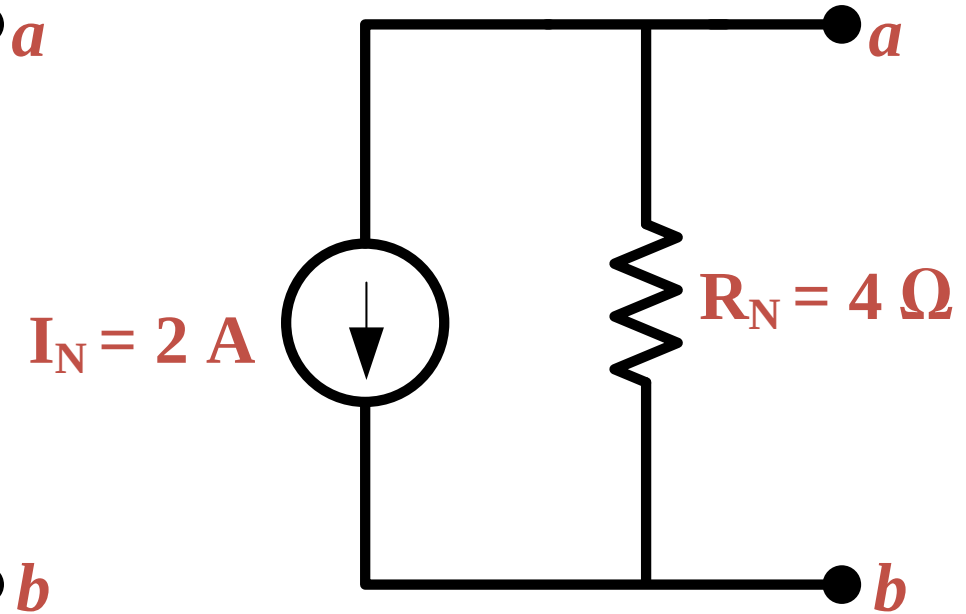
$$R_{Th} = R_N = 4\Omega$$

$$V_{Th} = -8V$$

$$I_N = -2A$$



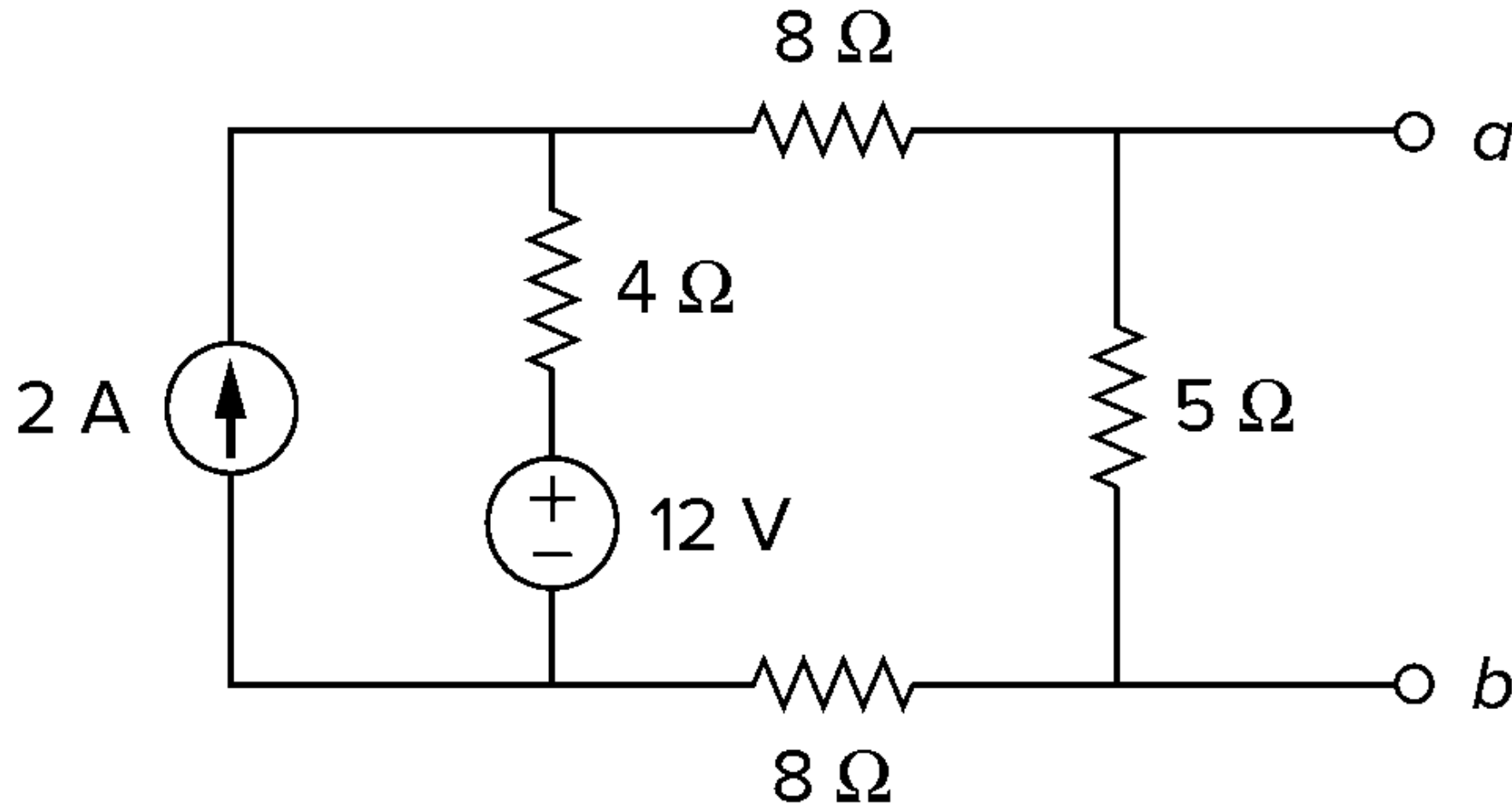
Thevenin
Equivalent Circuit



Norton Equivalent
Circuit

NUMERICAL

Q6. Find the Norton equivalent at terminals a - b of the circuit shown.



NUMERICAL

Soln:

Step 1: Finding R_N .

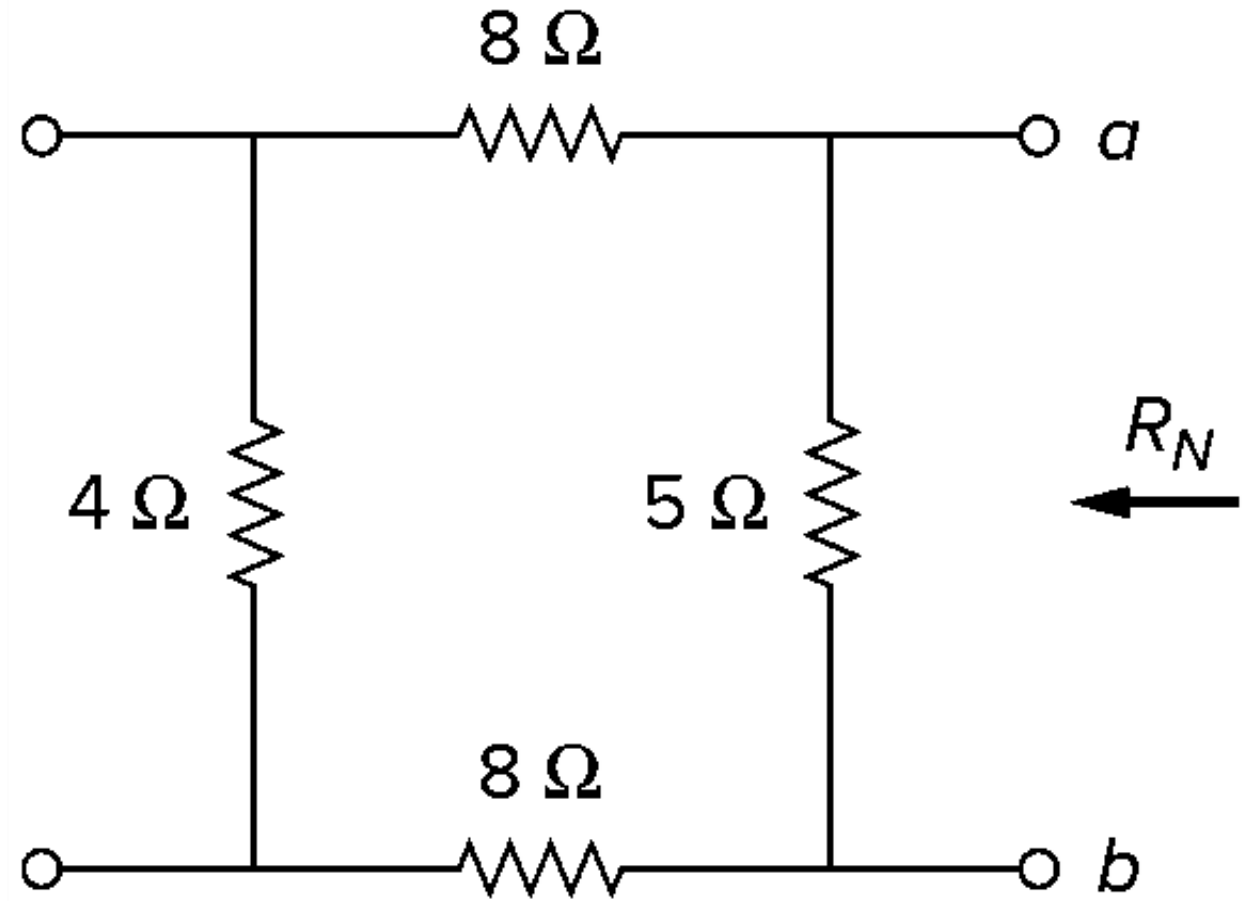
Set the independent sources to zero. This leads to the reduced circuit shown.

Thus,

$$R_N = 5 \parallel (8 + 4 + 8)$$

$$\Rightarrow R_N = 5 \parallel 20 = \frac{20 \times 5}{20 + 5}$$

$$\Rightarrow \boxed{R_N = 4\Omega}$$



NUMERICAL

Step 2: Finding I_N . Short-circuit terminals a and b .

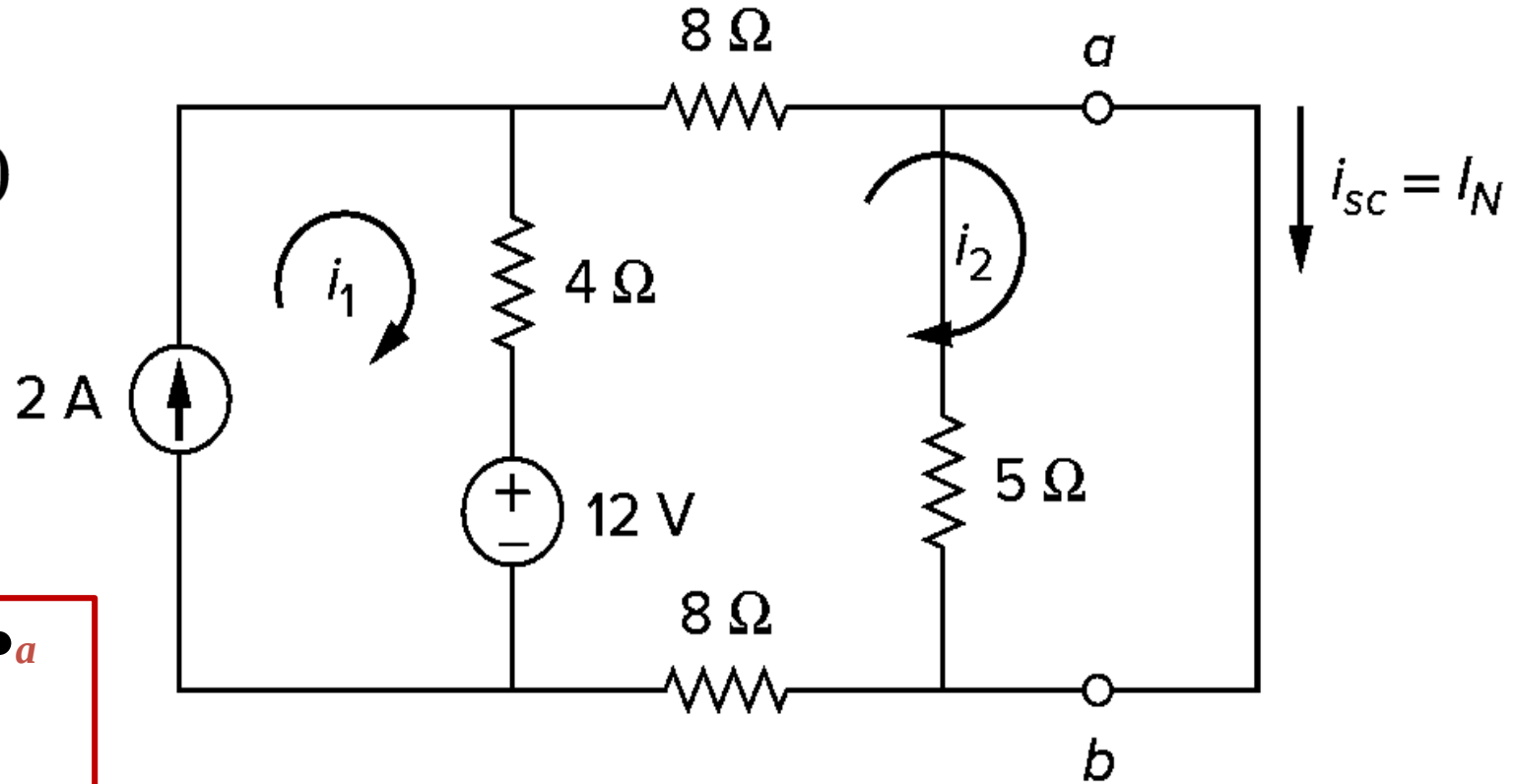
Ignore the $5\text{-}\Omega$ resistor because it has been short-circuited.

Applying mesh analysis,

$$i_1 = 2\text{A}; 20i_2 - 4i_1 - 12 = 0$$

On solving, we get

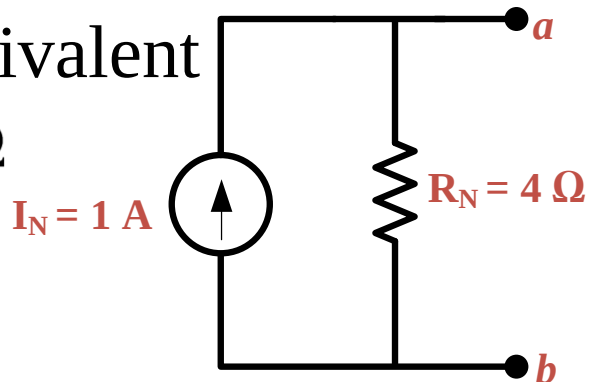
$$i_2 = \boxed{1\text{A}} = i_{sc} = \boxed{I_N}$$



Norton's Equivalent

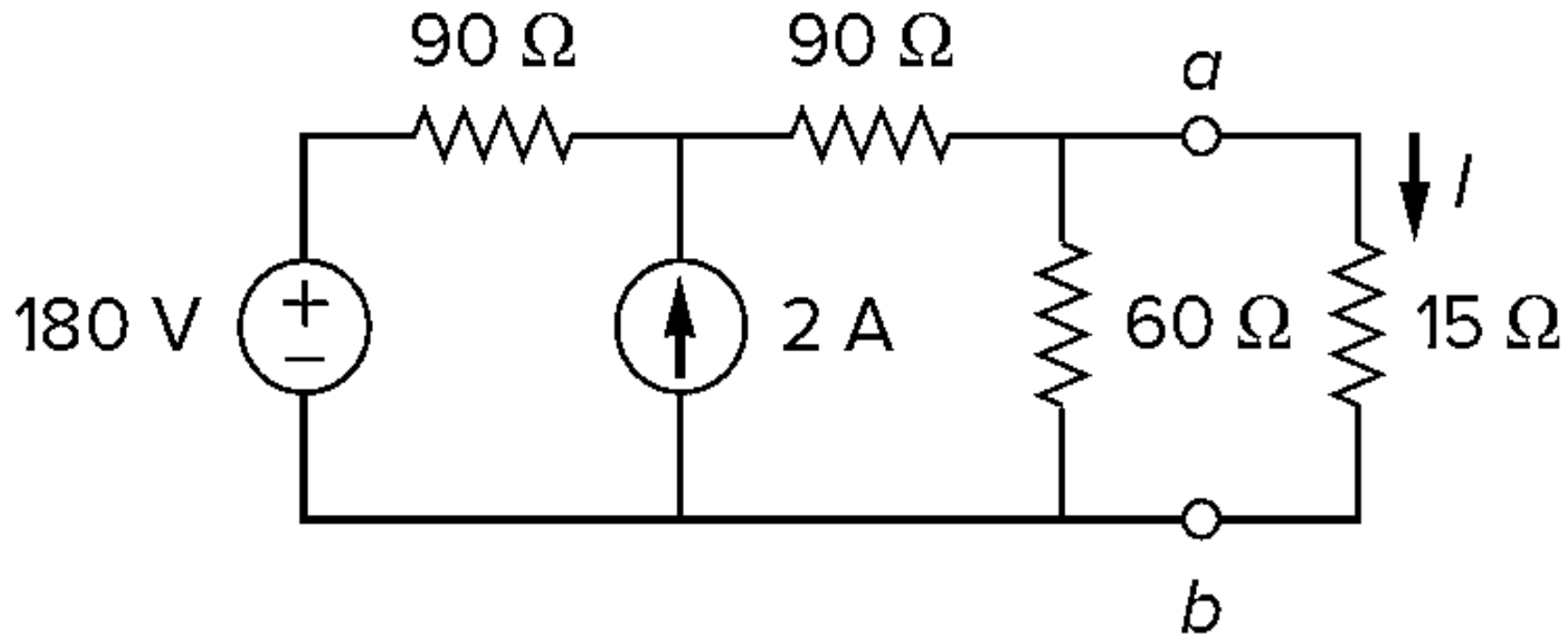
$$R_{Th} = R_N = 4\Omega$$

$$I_N = 1\text{A}$$



UNSOLVED NUMERICAL

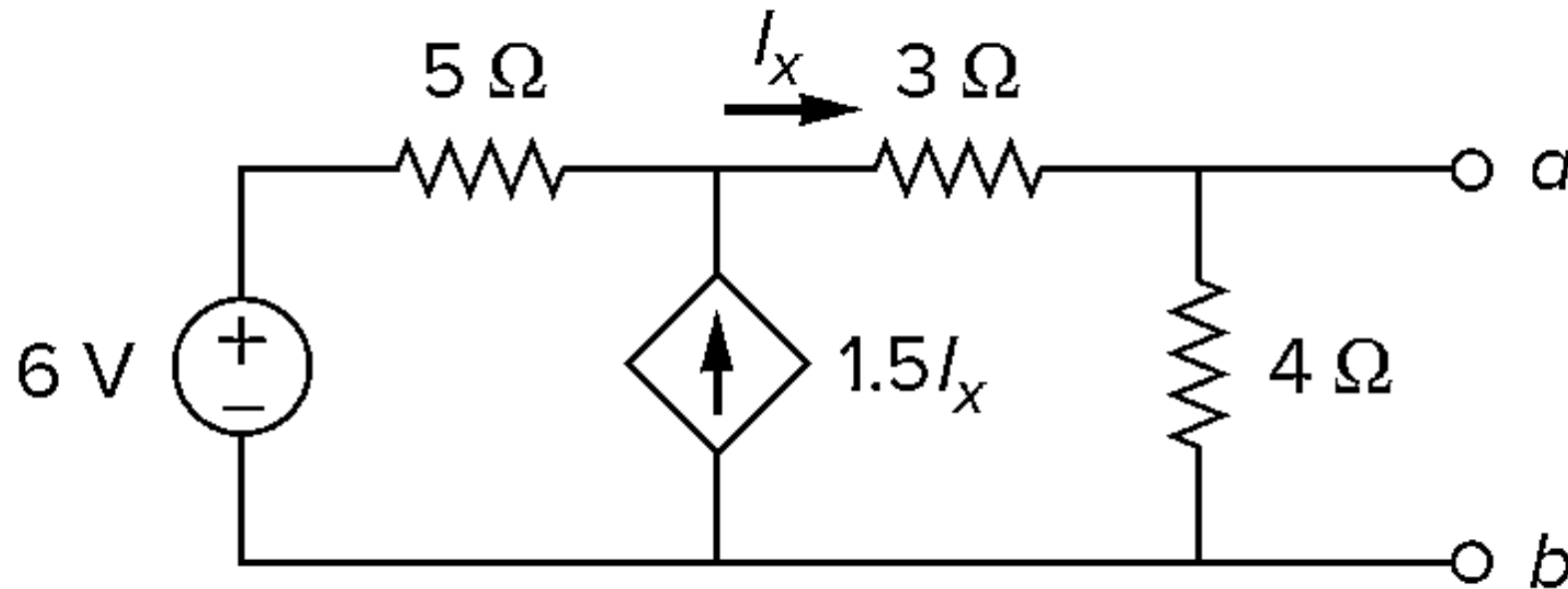
Q1. Using Thevenin's theorem, find the equivalent circuit to the left of the terminals in the circuit below. Then find I .



Ans: $V_{Th} = 90 \text{ V}$, $R_{Th} = 45 \text{ } \Omega$, $I = 1.5 \text{ A}$.

UNSOLVED NUMERICAL

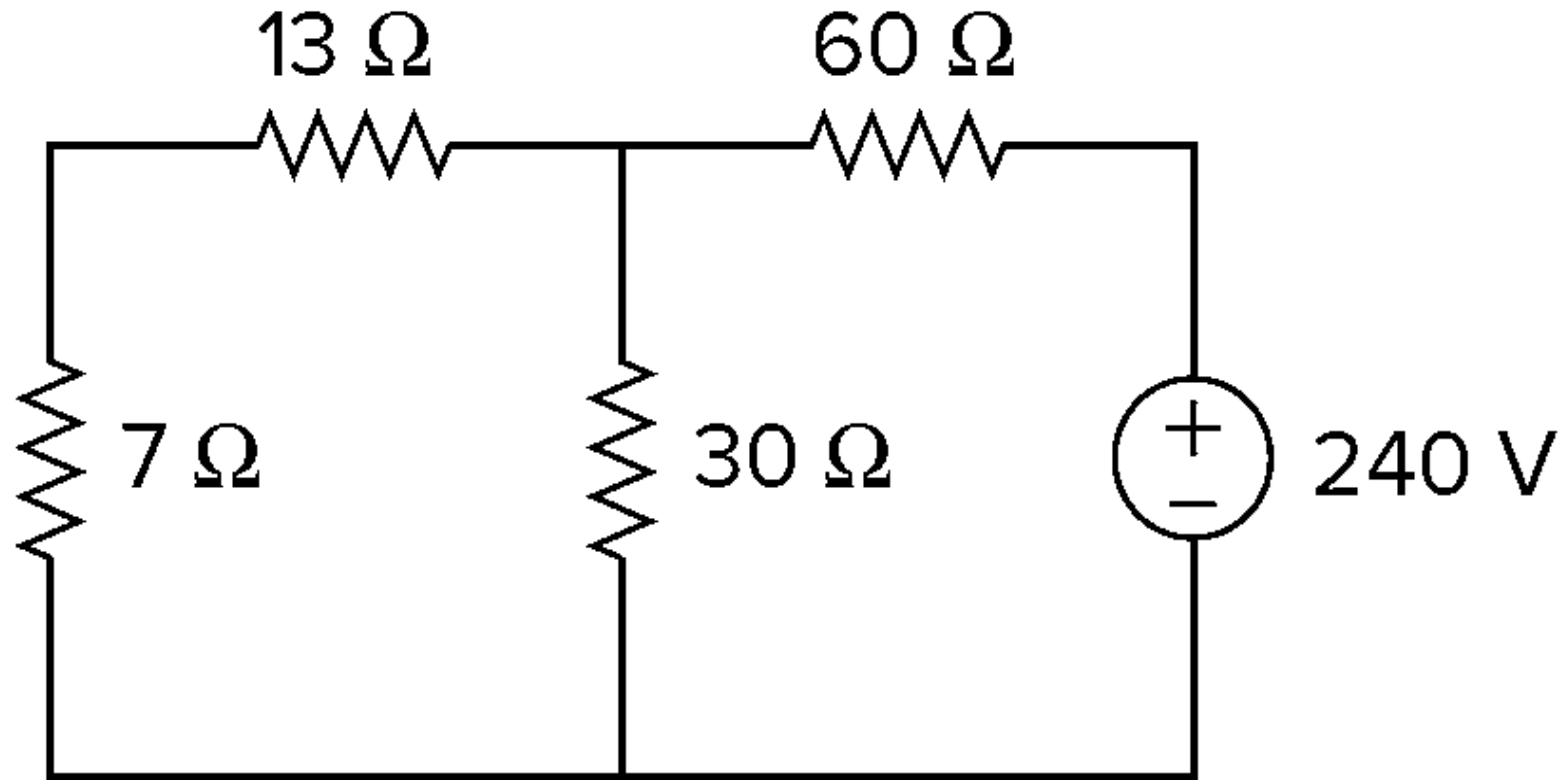
Q2. Find the Thevenin equivalent circuit of the circuit shown below to the left of the terminals.



Ans: $V_{Th} = 5.333 \text{ V}$, $R_{Th} = 444.4 \text{ m}\Omega$.

UNSOLVED NUMERICAL

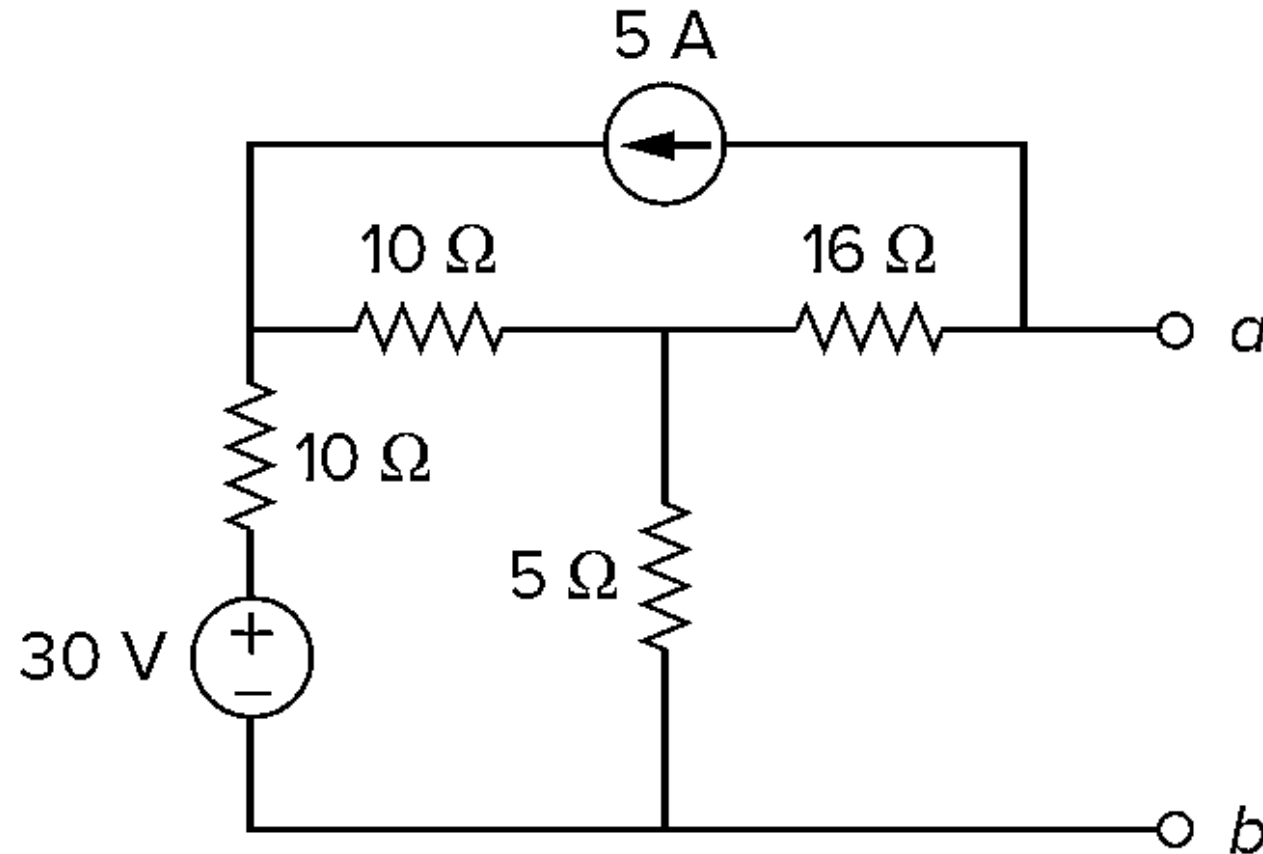
Q3. Determine the Thevenin equivalent circuit shown below, as seen by the 7Ω resistor. Then calculate the current flowing through the 7Ω resistor.



Ans: $V_{Th} = 80\text{ V}$, $R_{Th} = 9.069\Omega$.

UNSOLVED NUMERICAL

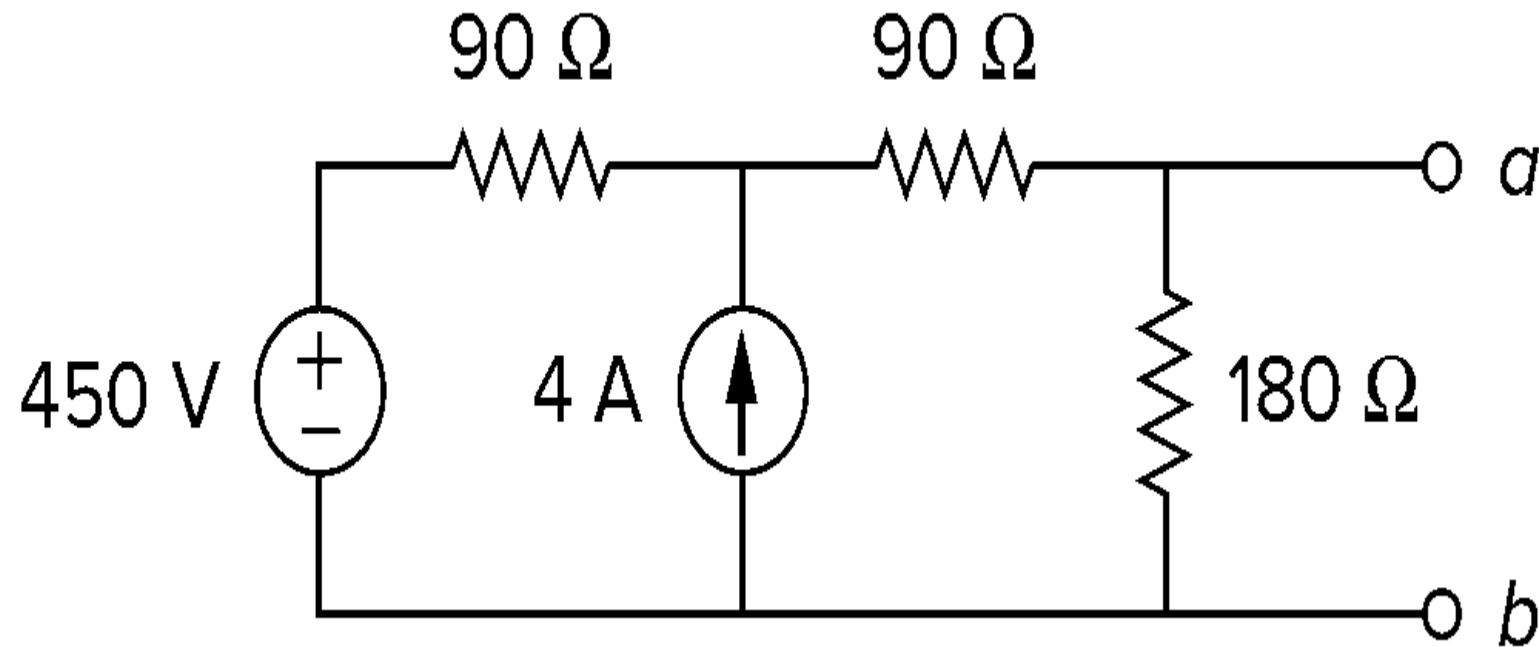
Q4. Obtain the Thevenin equivalent at terminals a-b of the circuit shown below.



Ans: $V_{Th} = 38 \text{ V}$, $R_{Th} = 20 \text{ } \Omega$.

UNSOLVED NUMERICAL

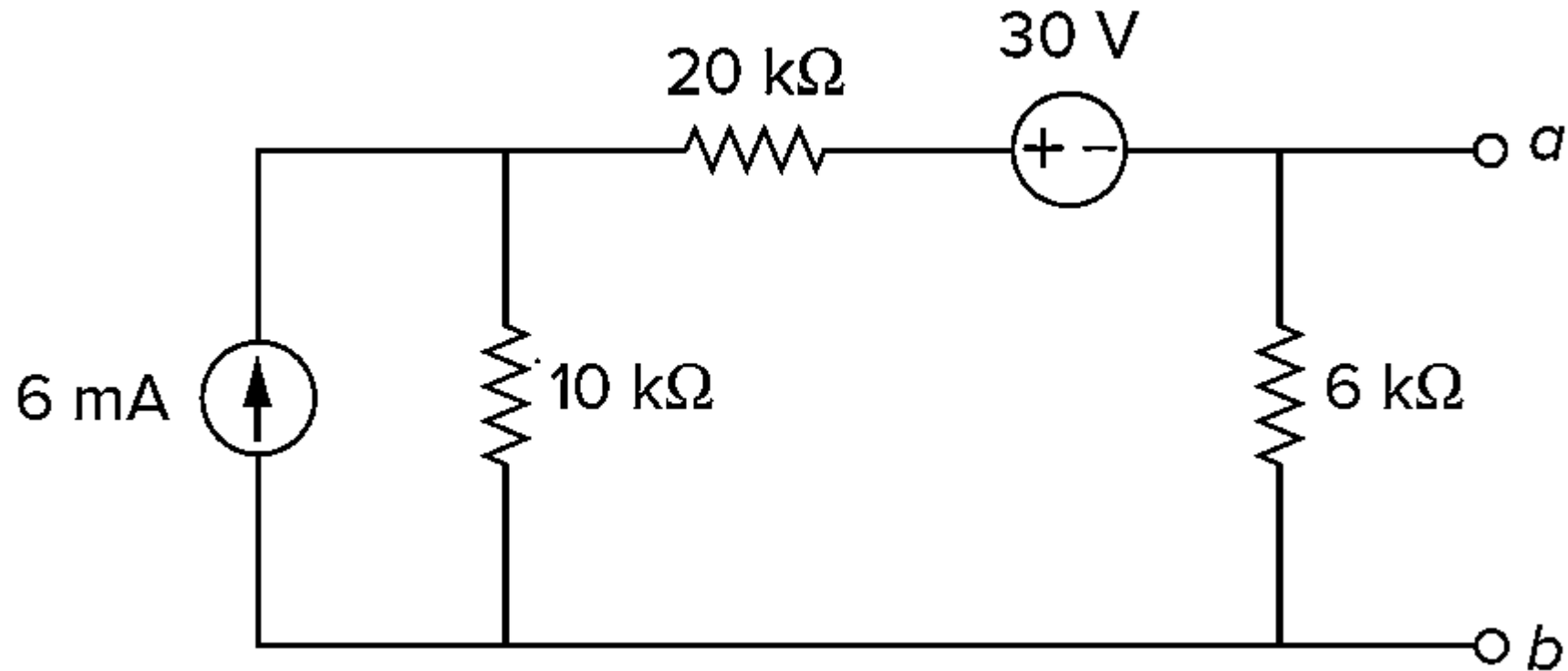
Q5. Find Norton equivalent resistance R_N and current I_N at terminals a-b of the circuit shown below.



Ans: $I_N = 4.5 \text{ A}$, $R_N = 90 \text{ } \Omega$.

UNSOLVED NUMERICAL

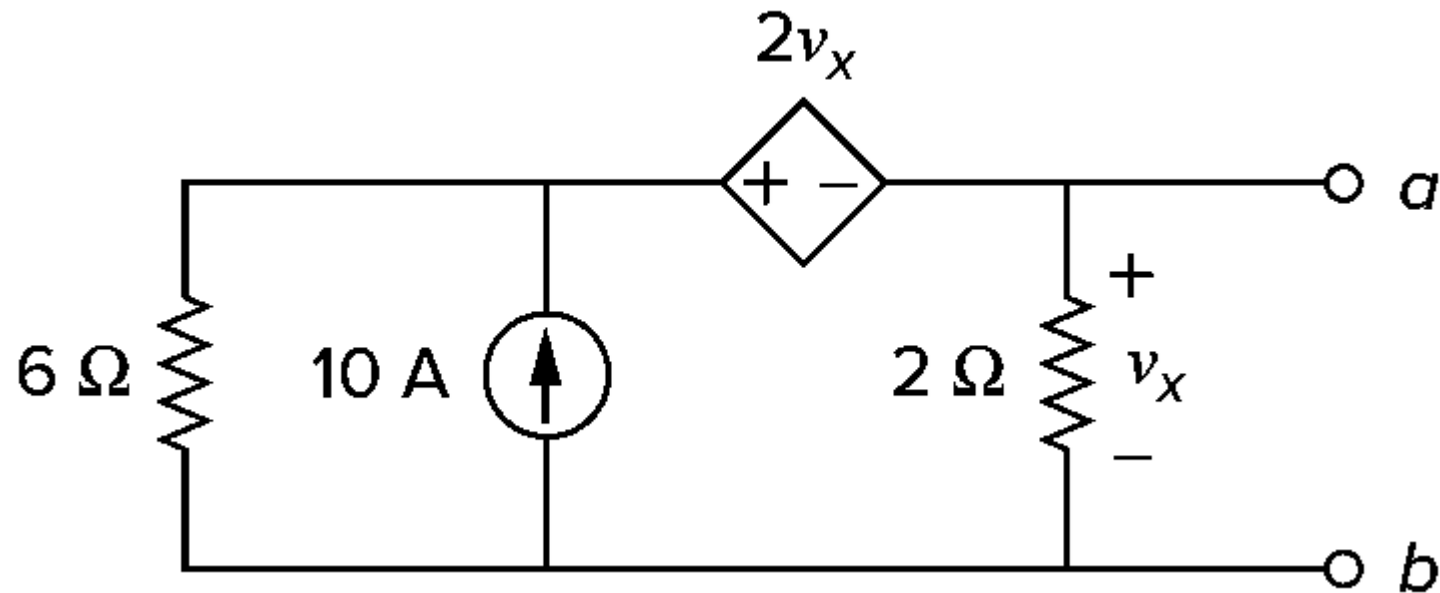
Q6. Find the Norton equivalent with respect to terminals a - b in the circuit shown below.



Ans: $I_N = 666.67\text{ mA}$, $R_N = 10\text{ }\Omega$.

UNSOLVED NUMERICAL

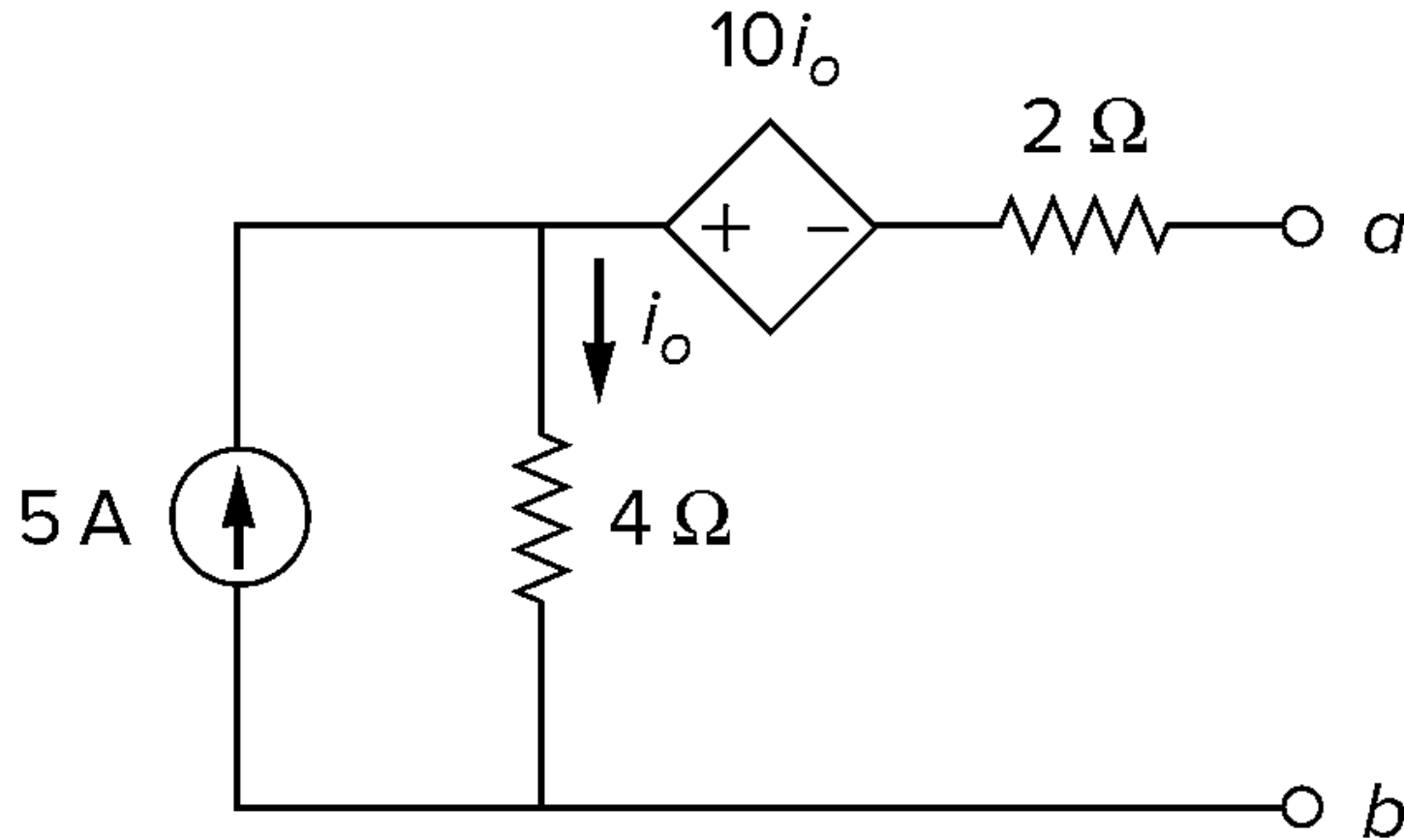
Q7. Find Norton equivalent resistance R_N and current I_N at terminals a-b of the circuit shown below.



Ans: $I_N = 10\text{ A}$, $R_N = 1\ \Omega$.

UNSOLVED NUMERICAL

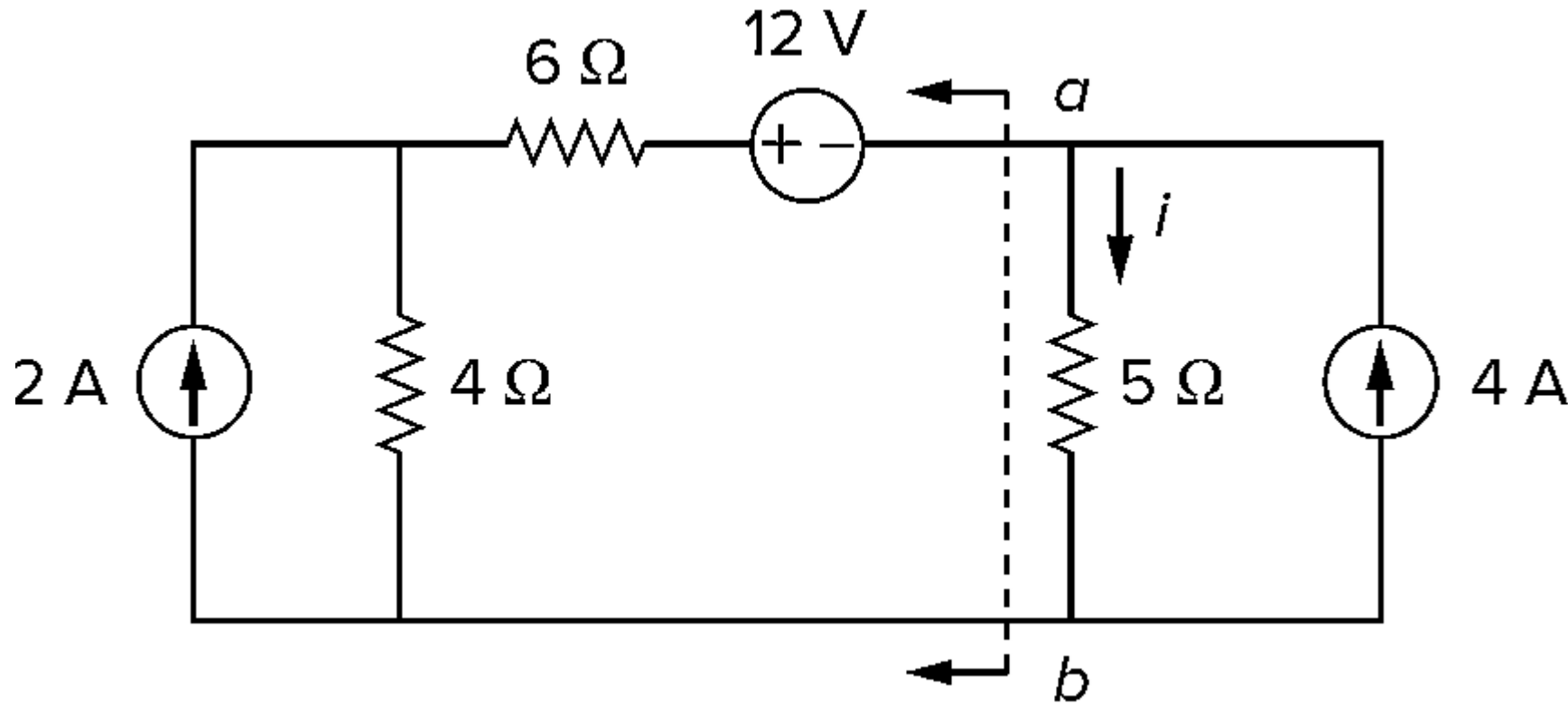
Q8. Determine the Norton equivalent at terminals a - b for the circuit below.



Ans: $I_N = 3 \text{ A}$, $R_N = -4 \text{ } \Omega$.

UNSOLVED NUMERICAL

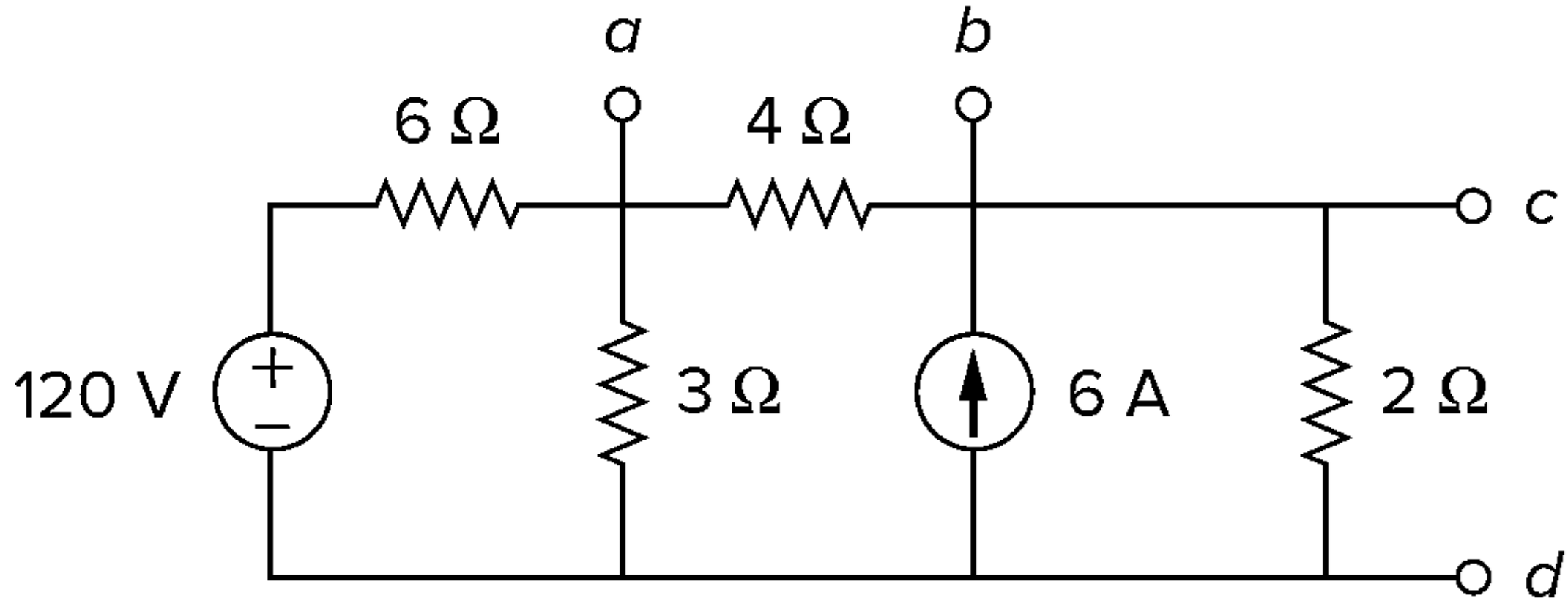
Q9. Obtain the Norton equivalent of the circuit in fig shown below, to the left of terminals a - b . Use the result to find current i .



Ans: $I_N = -0.4\text{ A}$, $R_N = 10\ \Omega$, $i = 2.4\text{ A}$.

UNSOLVED NUMERICAL

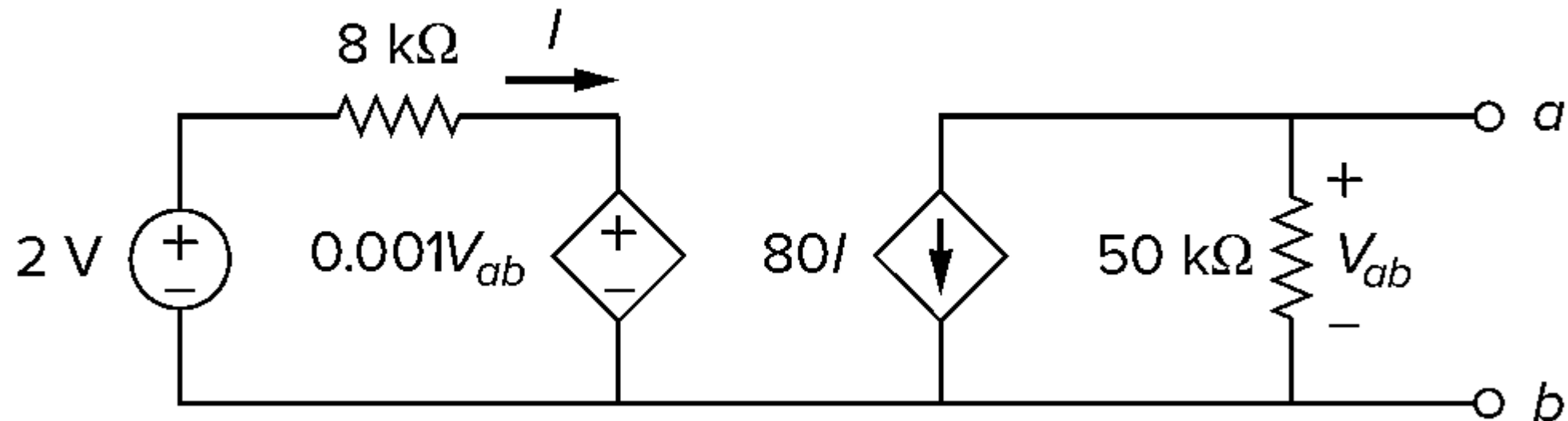
Q10. Given the circuit in Fig below, obtain the Norton equivalent as viewed from terminals: (a) a - b (b) c - d .



Ans: (a) $I_N = 7 \text{ A}$, $R_N = R_{Th} = 2 \Omega$, $V_{Th} = 14 \text{ V}$. (b) $I_N = 12.667 \text{ A}$, $R_N = R_{Th} = 1.5 \Omega$, $V_{Th} = 19 \text{ V}$.

UNSOLVED NUMERICAL

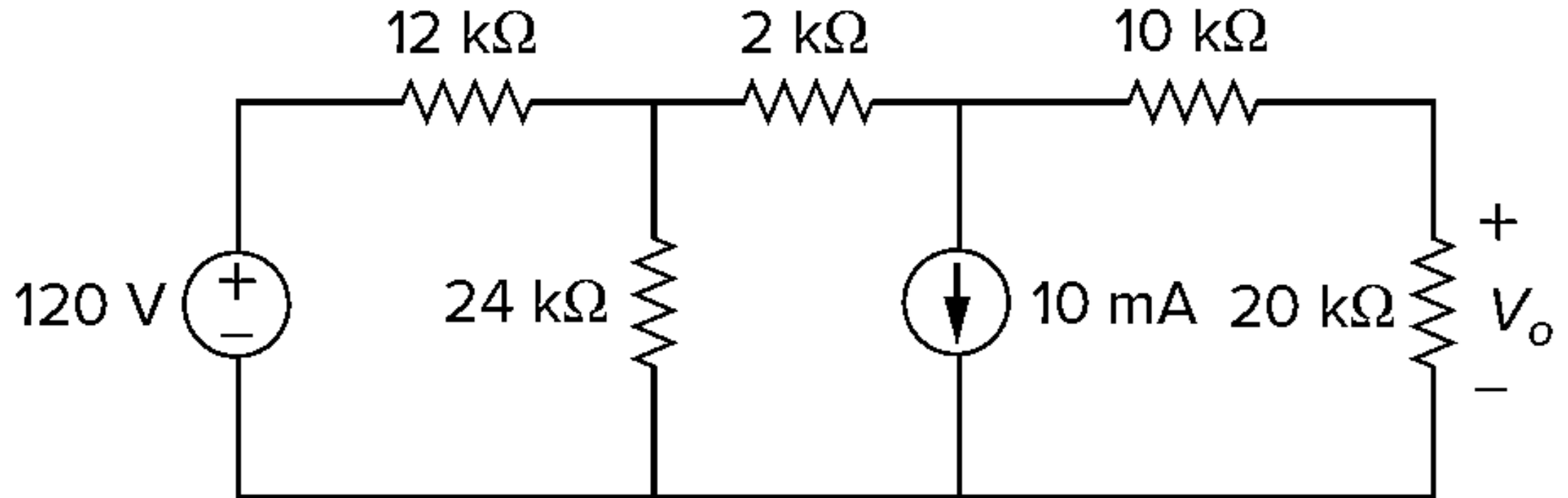
Q11. Obtain the Norton equivalent at terminals a - b of the circuit in Fig.



Ans: $I_N = -20\text{ mA}$, $R_N = 100\text{ k}\Omega$.

UNSOLVED NUMERICAL

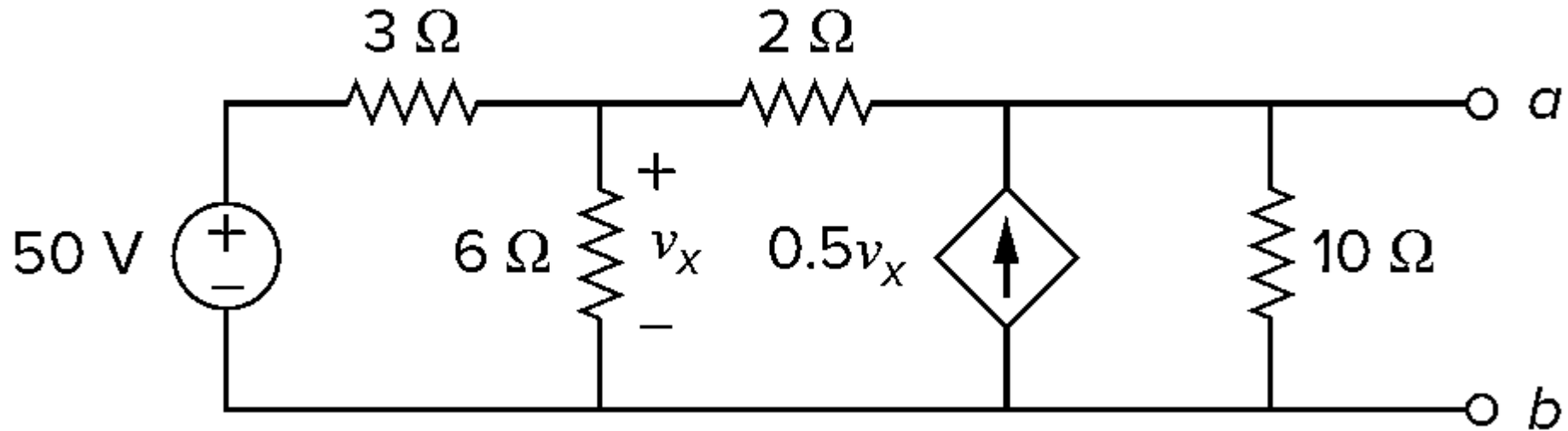
Q12. Use Norton's theorem to find V_o in the circuit of Fig.



Ans: $I_N = -1\text{ mA}$, $R_N = 20\text{ k}\Omega$.

UNSOLVED NUMERICAL

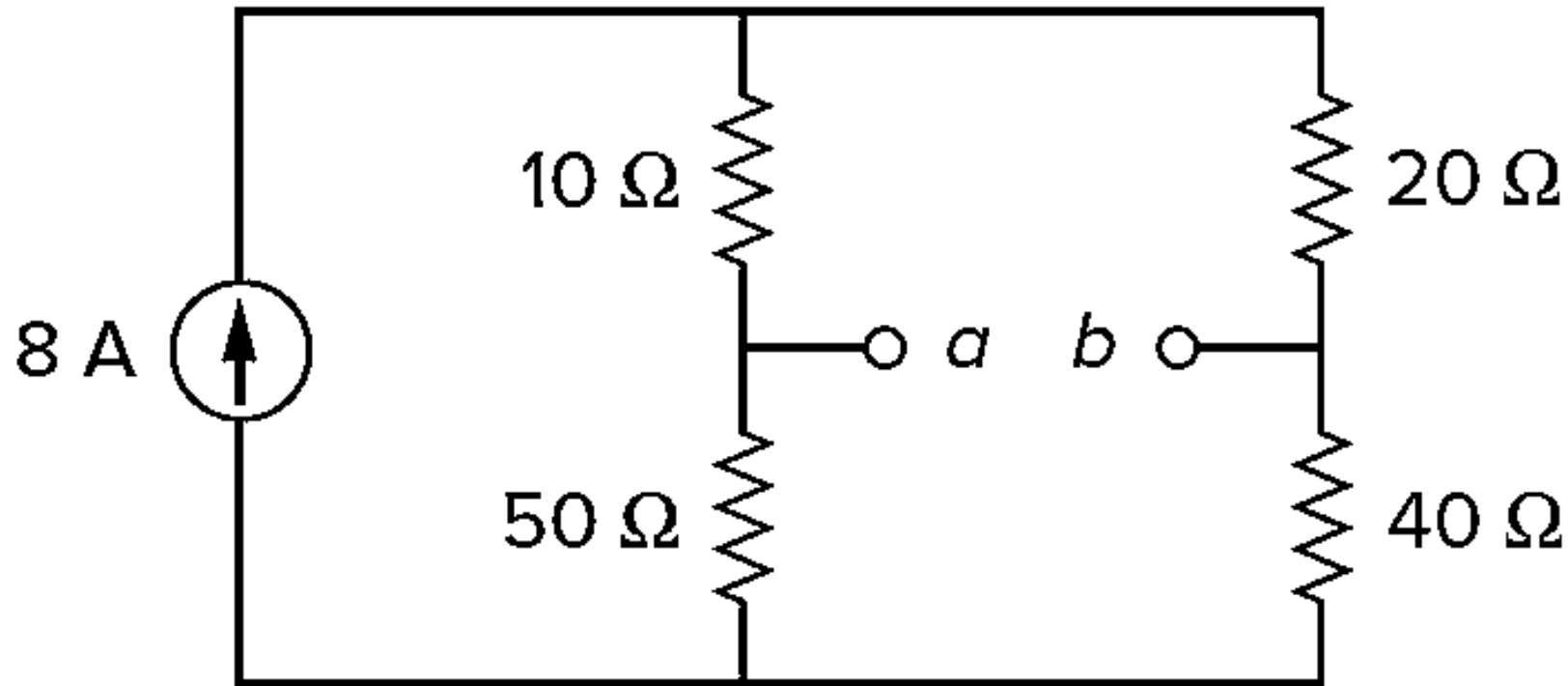
Q13. Obtain the Thevenin and Norton equivalent circuits at terminals a - b for the circuit in Fig.



Ans: (a) $I_N = 16.667$ A, $R_N = R_{Th} = 10$ Ω, $V_{Th} = 166.67$ V.

UNSOLVED NUMERICAL

Q14. Determine the Thevenin and Norton equivalents at terminals a - b of the circuit in Fig.



Ans: (a) $I_N = 1.7778$ A, $R_N = R_{Th} = 22.5$ Ω, $V_{Th} = 40$ V.

REFERENCES

1. Edward Hughes; John Hiley, Keith Brown, Ian McKenzie Smith, “*Electrical and Electronic Technology*”, 10th edition, Pearson Education Limited, Year: 2008.
2. Alexander, Charles K., and Sadiku, Matthew N. O., “*Fundamentals of Electric Circuits*”, 5th Ed, McGraw Hill, Indian Edition, 2013.